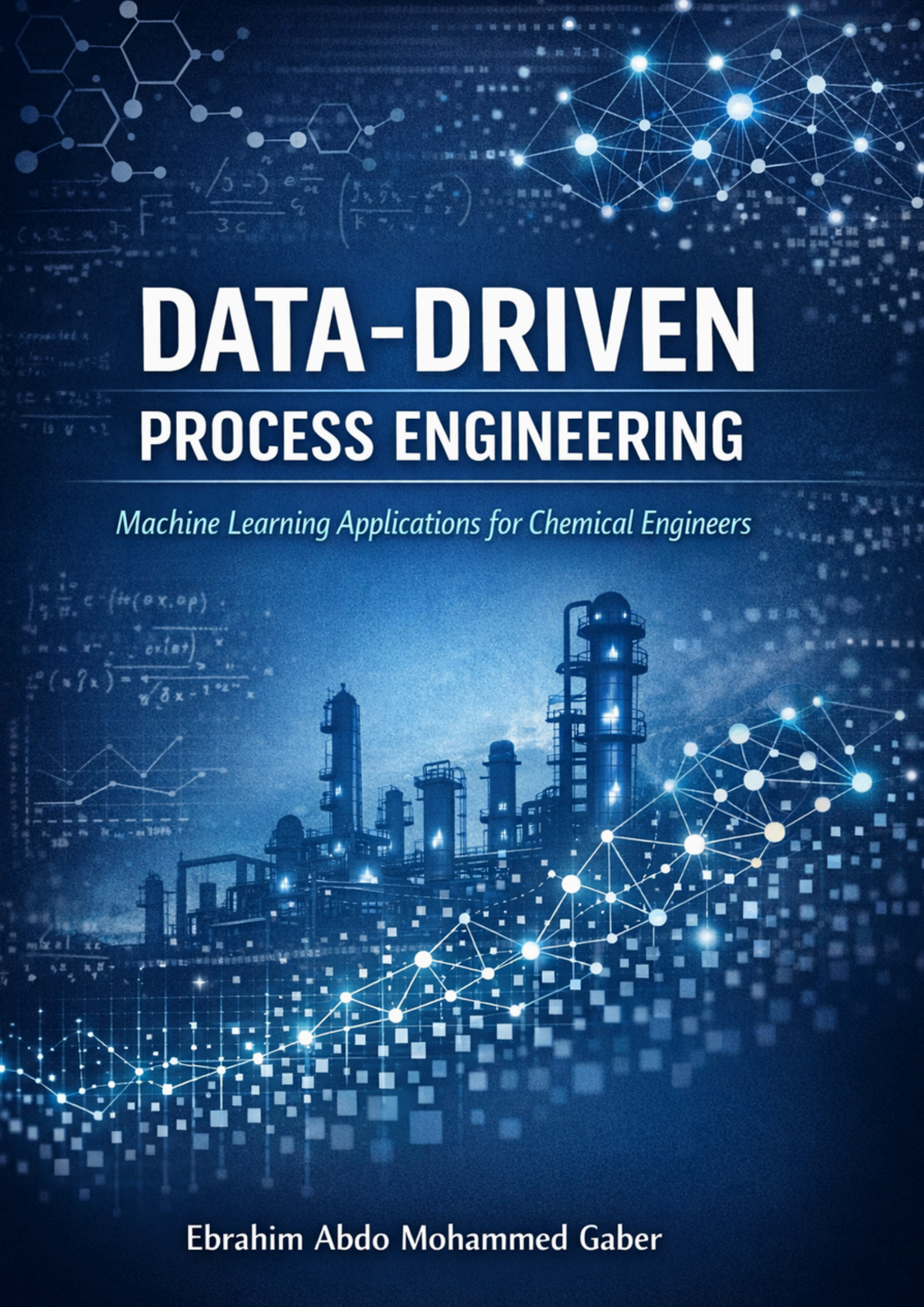




DATA-DRIVEN PROCESS ENGINEERING

Machine Learning Applications for Chemical Engineers

Ebrahim Abdo Mohammed Gaber

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Preface

Chemical engineering is undergoing a quiet revolution. For decades, the discipline has been built on first principles: thermodynamics, transport phenomena, reaction engineering, and process design. These foundations remain indispensable. However, modern industry now generates vast amounts of data—process variables, sensor streams, operational histories, and experimental results—that often remain underutilized. At the same time, machine learning has matured from an academic curiosity into a practical engineering tool. During my academic journey at Universiti Teknologi Malaysia (UTM), I was trained rigorously in process design, simulation, safety analysis, and environmental engineering. I worked on hydrothermal gasification for green hydrogen production, designed carbon capture systems, and conducted experimental research in biomass conversion. These experiences reinforced the power of classical chemical engineering methods. Yet, while working with experimental datasets and process simulations, I realized something important: Traditional models explain *why* systems behave the way they do. Machine learning models help us *predict* how they will behave next.

Rather than viewing these approaches as competing philosophies, I see them as complementary tools. This book was written to serve engineers who want to:

- Integrate machine learning into process design and optimization
- Use predictive modeling for sustainable chemical technologies
- Enhance R&D through intelligent data analysis
- Bridge academic research and industrial implementation

The goal is not to replace engineering fundamentals with algorithms. Instead, it is to empower engineers to augment their expertise with modern computational intelligence.

Whether you are:

- A graduate student entering research,
- A process engineer seeking optimization tools,
- An academic exploring data-driven methodologies,
- Or a consultant looking to provide high-impact analytical solutions,

this book is designed to provide both conceptual clarity and practical direction.

My specialization in machine learning and professional certification in data analytics reflect my belief that the future of chemical engineering lies at the intersection of sustainability, digitalization, and intelligent systems.

If this book inspires even one engineer to combine rigorous physical insight with modern data-driven tools, then it will have fulfilled its purpose.

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About the Author

Ebrahim Abdo Mohammed Gaber is a Chemical Engineer, R&D Specialist, and Data Analyst with a strong interdisciplinary focus on sustainable process engineering and machine learning applications in chemical systems. He graduated with a Bachelor of Engineering (Hons) in Chemical Engineering from Universiti Teknologi Malaysia (UTM) with a CGPA of 3.52/4.00

Ebrahim's technical foundation spans process simulation (Aspen HYSYS/Plus), equipment sizing, distillation design, process optimization, and environmental systems engineering. His academic and research experience includes hydrothermal gasification for green hydrogen production, carbon capture and utilization (CCU) plant design, effluent treatment systems, and environmental impact assessment projects. During his tenure as an Assistant Chemical Engineer (R&D) with the Chemical Reaction Engineering Group (CREG) at UTM, he designed and executed biomass gasification experiments, performed gas chromatography analysis, and applied statistical modeling using Response Surface Methodology (RSM). His work reflects a commitment to advancing green hydrogen technologies and sustainable chemical solutions. Recognizing the transformative role of artificial intelligence in engineering, Ebrahim pursued advanced training in data science and machine learning. He completed the Machine Learning Specialization and Data Analytics Specialization from DeepLearning.AI, along with additional certifications in process safety, experimental design, and advanced Excel analytics. His combined expertise in chemical engineering, machine learning, and data analytics positions him at the intersection of process innovation and intelligent systems design. Through this book, he aims to bridge traditional chemical engineering principles with modern data-driven methodologies, empowering engineers to build smarter, safer, and more sustainable industrial systems.

0.1 Introduction

Chemical engineering has historically been grounded in first-principles modeling. The discipline was built upon conservation laws, thermodynamics, transport phenomena, reaction kinetics, and rigorous mathematical analysis. For decades, these principles enabled engineers to design reactors, separation systems, and control architectures with remarkable predictive capability. However, the industrial landscape has evolved. Modern chemical processes generate unprecedented volumes of data through distributed control systems, advanced sensors, online analyzers, and digital manufacturing platforms. This transformation has created both an opportunity and a challenge: how to extract meaningful knowledge from complex, high-dimensional process data while maintaining the physical rigor that defines engineering practice.

Machine learning offers a complementary modeling paradigm. Rather than deriving governing equations exclusively from mechanistic principles, machine learning constructs predictive relationships directly from data. In doing so, it provides tools capable of handling nonlinear behavior, multivariable interactions, noisy measurements, and complex patterns that may be analytically

intractable. Yet, without careful integration into engineering reasoning, machine learning risks becoming a purely computational exercise detached from physical understanding.

This book was written to bridge that gap.

The objective is not merely to introduce algorithms, nor to provide superficial tutorials in programming. Instead, this text develops machine learning from an engineering perspective—anchored in chemical process understanding, mathematical clarity, and practical applicability. Every method presented is interpreted through the lens of chemical engineering: how it relates to classical modeling, where its assumptions hold, and where its limitations must be respected.

The structure of the book reflects a deliberate progression.

Part I establishes foundational concepts, introducing machine learning terminology, data handling, and the computational tools required for implementation. The emphasis is placed on understanding data as a representation of physical systems rather than as abstract numerical arrays

Part II develops supervised learning methods for regression and classification, framing them as extensions of familiar engineering tasks such as parameter estimation, performance prediction, and operational stability analysis.

Part III addresses unsupervised learning and dimensionality reduction, providing tools for interpreting complex multivariable datasets, identifying operational regimes, and extracting dominant process patterns.

Part IV advances toward nonlinear modeling, time-dependent systems, validation methodologies, and integration strategies—areas where machine learning intersects most directly with industrial deployment and decision support.

Part V provides technical appendices to support reproducibility, implementation, and continued study.

Throughout the text, several principles are maintained:

- Concepts are introduced before mathematics.
- Mathematics is presented to clarify meaning, not to obscure it.
- Algorithms are interpreted physically whenever possible.
- Synthetic datasets are constructed to reflect realistic chemical process behavior.
- Model validation, uncertainty, and limitations are treated as central—not peripheral—topics.

The intended audience includes chemical engineers with no prior background in machine learning, graduate students seeking to expand their modeling toolkit, and practicing engineers transitioning into data-driven roles. The goal is to provide both conceptual depth and practical competence, enabling readers to develop, evaluate, and responsibly deploy machine learning models in chemical engineering contexts.

Machine learning does not replace first-principles modeling. Rather, it expands the engineer’s analytical repertoire. When used thoughtfully, it can enhance predictive accuracy, reveal hidden

process structure, support optimization, and improve operational resilience. When used carelessly, it can produce misleading results divorced from physical reality. The distinction lies in understanding.

This book seeks to cultivate that understanding.

It invites the reader to approach machine learning not as a collection of algorithms, but as an engineering discipline grounded in rigor, interpretation, and responsible application.

1 Chapter 1: Introduction to Machine Learning in Chemical Engineering

1.1 Introduction

Machine learning (ML) has emerged as a transformative tool in chemical engineering, enabling engineers to extract actionable insights from complex, multivariate datasets. Unlike traditional first-principles modeling, which relies on explicit physical laws, ML models can identify patterns, correlations, and nonlinear relationships directly from data. This capability is particularly valuable in processes where mechanistic understanding is limited, experimental data are abundant, or process dynamics are highly nonlinear.

This chapter introduces the fundamental concepts of machine learning, designated specifically for chemical engineers. It establishes a foundation for subsequent chapters that delve into supervised, unsupervised, and advanced machine learning methods applied to chemical processes.

1.2 Motivation

The modern chemical engineer is increasingly faced with data-rich environments:

- **Process optimization:** Large-scale plants produce thousands of measurements per second from sensors and control systems.
- **Predictive maintenance:** Anticipating equipment failures can significantly reduce downtime and operational costs.
- **Data-driven control:** Machine learning enables the design of adaptive controllers that respond to changing process conditions.

Machine learning provides a systematic framework for leveraging this data to improve process understanding, design, and operation.

1.3 Engineering Relevance

Machine learning complements classical chemical engineering approaches:

- **Reactor modeling:** Predicting yields and selectivity from multiple interacting process variables.
- **Separation processes:** Estimating performance metrics without explicitly solving complex partial differential equations or relying solely on empirical correlations.
- **Energy efficiency:** Identifying optimal operating conditions from historical plant data.

By integrating machine learning with engineering intuition, practitioners can generate models that are not only accurate but also interpretable and aligned with underlying physical principles.

1.4 Conceptual Foundations

1.4.1 What is Machine Learning?

Machine learning is the study of algorithms that improve their performance at a given task over time with experience, typically by learning patterns from data.

Formally, a machine learning model seeks to approximate an unknown functional relationship:

$$\mathbf{Y} = f(\mathbf{X}) + \varepsilon$$

where:

- $\mathbf{X} \in \mathbb{R}^{n \times p}$ represents the matrix of input variables,
- $\mathbf{Y} \in \mathbb{R}^{n \times q}$ represents the output variables,
- $f(\cdot)$ denotes an unknown, potentially nonlinear mapping,
- ε represents measurement noise or modeling error.

1.4.2 Key Categories of Machine Learning

- **Supervised learning:** Learn a mapping from inputs $\mathbf{X}$ to outputs $\mathbf{Y}$ using labeled data.

Example: Predicting reactor yield Y as a function of temperature T , pressure P , and feed composition $\mathbf{z}$:

$$Y = f(T, P, \mathbf{z})$$

- **Unsupervised learning:** Identify structure or patterns in unlabeled data without predefined output variables.

Example: Clustering operating regimes of a distillation column based on multivariate sensor data.

- **Reinforcement learning:** Learn optimal decision policies through trial-and-error interaction with a dynamic system.

Example: Optimizing a multistage separation process via sequential control actions to maximize product purity or minimize energy consumption.

1.5 Chemical Engineering Perspective

In chemical engineering applications, data typically originate from laboratory experiments, process simulations, or industrial plant sensors. Machine learning models provide:

- **Nonlinear mapping:** Capture complex interactions among variables that are difficult to model using first-principles alone.
- **Dimensionality reduction:** Transform high-dimensional sensor data into lower-dimensional representations suitable for visualization or control.
- **Predictive capability:** Forecast process behavior under untested or hypothetical operating conditions.

Importantly, machine learning does not replace first-principles models. Instead, it augments them, enabling hybrid modeling approaches that combine mechanistic understanding with data-driven inference.

1.6 Python Implementation: Exploring Process Data

We begin by generating a synthetic dataset representing a chemical reactor, including temperature (K), pressure (bar), feed concentration (mol L^{-1}), and product yield (%).

```
import numpy as np
import pandas as pd

# Set random seed for reproducibility
np.random.seed(42)

# Generate synthetic process data
n_samples = 100
temperature = np.random.normal(loc=350, scale=10, size=n_samples) # K
pressure = np.random.normal(loc=5, scale=0.5, size=n_samples) # bar
feed_conc = np.random.normal(loc=2, scale=0.2, size=n_samples) # mol/L

# Simulate yield using a nonlinear relationship
yield_percent = 70 + 0.1*(temperature-350) + 2*(pressure-5) - 5*(feed_conc-2)**2
yield_percent += np.random.normal(0, 1, size=n_samples) # add noise

# Create DataFrame
data = pd.DataFrame({
    'Temperature_K': temperature,
    'Pressure_bar': pressure,
    'Feed_Conc_mol_L': feed_conc,
    'Yield_percent': yield_percent
})
```

```
data.head()
```

#This DataFrame provides a realistic, physically meaningful dataset to explore in subsequent c

	Temperature_K	Pressure_bar	Feed_Conc_mol_L	Yield_percent
0	354.967142	4.292315	2.071557	68.226746
1	348.617357	4.789677	2.112157	68.818013
2	356.476885	4.828643	2.216610	70.817668
3	365.230299	4.598861	2.210760	71.109023
4	347.658466	4.919357	1.724466	69.204065

1.7 Engineering Case Study: Predicting Reactor Yield

1.7.1 Problem Formulation

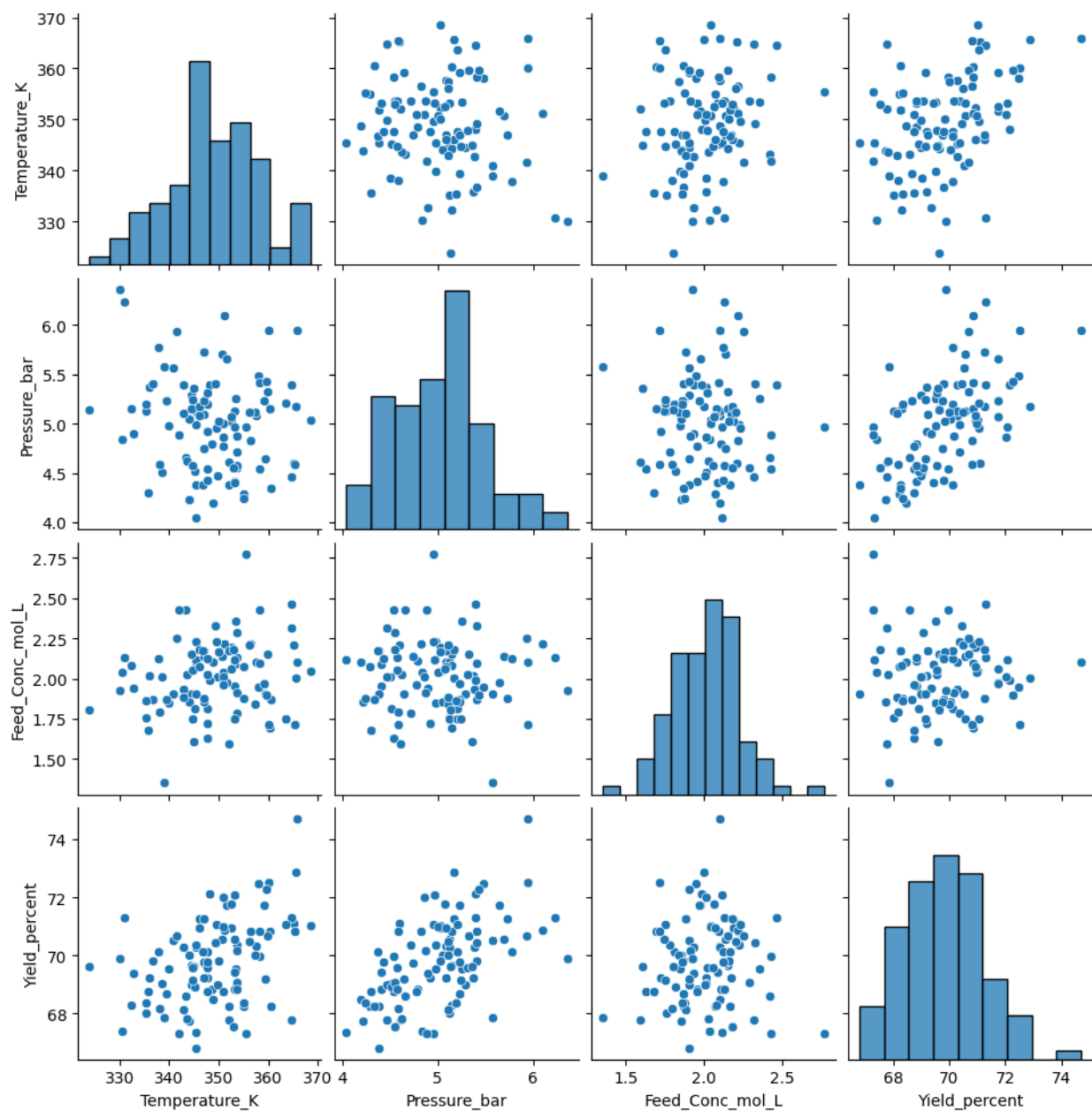
The objective is to predict the reactor yield (%) based on operating conditions. This constitutes a **supervised regression problem**, where the input variables are temperature, pressure, and feed concentration, and the output variable is the reactor yield.

- steps**
1. Generate or collect process data (done above).
 2. Explore data distributions and correlations.
 3. Apply regression or other ML techniques to model the output.

```
# Exploratory Data Analysis
import matplotlib.pyplot as plt
import seaborn as sns

# Pairplot to visualize relationships
sns.pairplot(data)
plt.show()

# Correlation matrix
corr_matrix = data.corr()
print(corr_matrix)
```



	Temperature_K	Pressure_bar	Feed_Conc_mol_L	Yield_percent
Temperature_K	1.000000	-0.136422	0.190840	0.411537
Pressure_bar	-0.136422	1.000000	-0.036632	0.556666
Feed_Conc_mol_L	0.190840	-0.036632	1.000000	0.047596
Yield_percent	0.411537	0.556666	0.047596	1.000000

Observations from this analysis guide feature selection, model choice, and interpretation, ensuring alignment with engineering intuition.

1.8 Chapter Summary

- Machine learning is a powerful tool to complement classical chemical engineering modeling.
- Supervised, unsupervised, and reinforcement learning each have distinct applications in process engineering.
- Conceptual understanding and engineering intuition are crucial to interpret ML outputs correctly.
- Python provides a flexible environment for data handling, visualization, and modeling.
- Synthetic, physically meaningful datasets allow experimentation and practice without relying on proprietary plant data.
- This chapter sets the foundation for regression and classification techniques, which will be explored in the next chapters, providing practical tools to model chemical processes with ML.

2 Chapter 2: Python and Data Handling for Engineers

2.1 Introduction

The effectiveness of machine learning in chemical engineering depends critically on how data are represented, processed, and interpreted before any model is constructed. Python has emerged as the dominant language for scientific computing and data-driven modeling due to its clarity, extensibility, and strong ecosystem of numerical libraries. For chemical engineers, Python serves not as a replacement for physical modeling, but as a computational environment that enables structured handling of experimental, simulation, and plant data.

This chapter introduces Python from an engineering perspective, emphasizing data structures and workflows that align naturally with chemical process variables. The objective is not to teach programming for its own sake, but to establish the computational literacy required for reliable machine learning modeling.

2.2 Refresher on Python for Scientific Computing

2.2.1 Variables, Arrays, and Data Types

In Python, variables are symbolic references to objects stored in memory. Unlike strongly typed engineering software, Python variables do not require explicit type declarations. While this flexibility accelerates development, it also places responsibility on the engineer to maintain clarity and consistency.

From a chemical engineering standpoint, numerical data typically fall into three categories:

- **Scalars:** single values such as temperature, pressure, or flow rate
- **Vectors:** measurements across time, space, or experimental runs
- **Matrices:** multivariable datasets representing multiple operating conditions

The NumPy library provides array-based data structures that efficiently represent these quantities.

```
import numpy as np

# Scalar
temperature = 350.0 # K

# Vector: temperature measurements
```

```

temperature_profile = np.array([345, 348, 350, 352, 355])

# Matrix: operating conditions (rows = experiments, columns = variables)
operating_conditions = np.array([
    [350, 5.0, 2.0],
    [360, 5.2, 2.1],
    [340, 4.8, 1.9]
])

temperature, temperature_profile, operating_conditions

```

```

(350.0,
 array([345, 348, 350, 352, 355]),
 array([[350. ,  5. ,  2. ],
        [360. ,  5.2,  2.1],
        [340. ,  4.8,  1.9]]))

```

Each column of the matrix may correspond to a process variable such as temperature, pressure, or feed concentration, mirroring the structure of experimental datasets.

2.3 Functions and Modular Code

Chemical engineering models often involve repeated calculations: evaluating thermodynamic properties, computing conversion, or estimating performance metrics. Encapsulating these operations in functions improves readability and reproducibility.

```

def reactor_yield(temperature, pressure, feed_conc):
    """
    Estimate reactor yield (%) based on operating conditions.
    """
    yield_estimate = (
        70
        + 0.1 * (temperature - 350)
        + 2.0 * (pressure - 5.0)
        - 5.0 * (feed_conc - 2.0) ** 2
    )
    return yield_estimate

reactor_yield(350, 5.0, 2.0)

```

```
70.0
```

Functions serve as computational analogs to algebraic correlations commonly used in process design.

2.4 Data Handling with NumPy and Pandas

2.4.1 Creating and Manipulating Arrays and DataFrames

While NumPy arrays are ideal for numerical operations, real engineering datasets require labeled variables and flexible indexing. Pandas provides the **DataFrame**, a table-like structure that closely resembles experimental spreadsheets and process historians.

```
import pandas as pd

data = pd.DataFrame({
    "Temperature_K": [345, 350, 355, 360],
    "Pressure_bar": [4.8, 5.0, 5.2, 5.1],
    "Feed_Conc_mol_L": [1.9, 2.0, 2.1, 2.0],
    "Yield_percent": [68, 71, 73, 72]
})

data
```

	Temperature_K	Pressure_bar	Feed_Conc_mol_L	Yield_percent
0	345	4.8	1.9	68
1	350	5.0	2.0	71
2	355	5.2	2.1	73
3	360	5.1	2.0	72

Each column corresponds to a physical variable, preserving engineering meaning and unit awareness.

2.5 Missing Data Handling, Scaling, and Normalization

In industrial environments, missing or corrupted sensor data are unavoidable. Ignoring these issues can lead to biased or unstable models.

```
# Introduce missing data
data.loc[2, "Pressure_bar"] = np.nan
data
```

	Temperature_K	Pressure_bar	Feed_Conc_mol_L	Yield_percent
0	345	4.8	1.9	68
1	350	5.0	2.0	71
2	355	NaN	2.1	73
3	360	5.1	2.0	72

Handling missing values requires engineering judgment:

```
# Replace missing pressure with mean value
data["Pressure_bar"] = data["Pressure_bar"].fillna(data["Pressure_bar"].mean())
data
```

	Temperature_K	Pressure_bar	Feed_Conc_mol_L	Yield_percent
0	345	4.800000	1.9	68
1	350	5.000000	2.0	71
2	355	4.966667	2.1	73
3	360	5.100000	2.0	72

Scaling is equally important, as ML algorithms are sensitive to variable magnitudes.

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
scaled_data = scaler.fit_transform(data[["Temperature_K", "Pressure_bar", "Feed_Conc_mol_L"]])

scaled_data
```

```
array([[ -1.34164079,  -1.5430335 ,  -1.41421356],
       [-0.4472136 ,   0.3086067 ,   0.          ],
       [ 0.4472136 ,   0.          ,   1.41421356],
       [ 1.34164079,   1.2344268 ,   0.          ]])
```

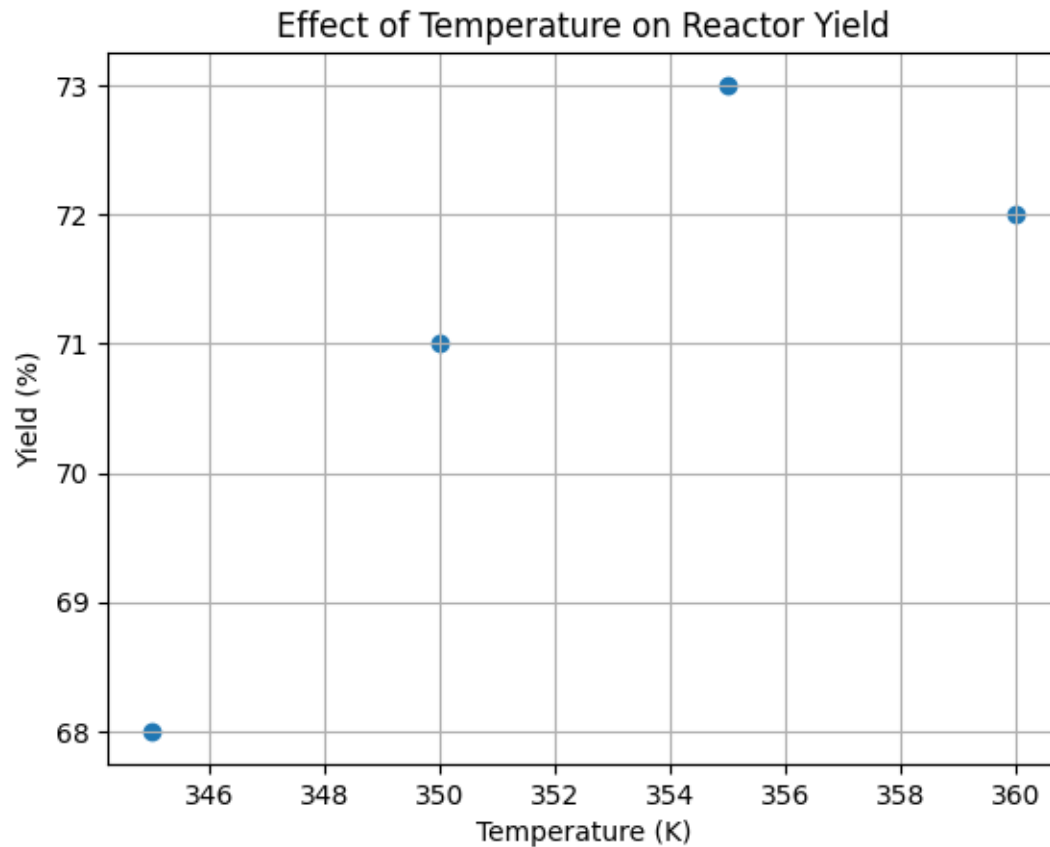
Scaling ensures that no single variable dominates due to units rather than physical importance.

2.6 Visualization of Process Data

Visualization is a diagnostic tool, not merely a presentation aid. For engineers, plots reveal correlations, anomalies, and operating regimes.

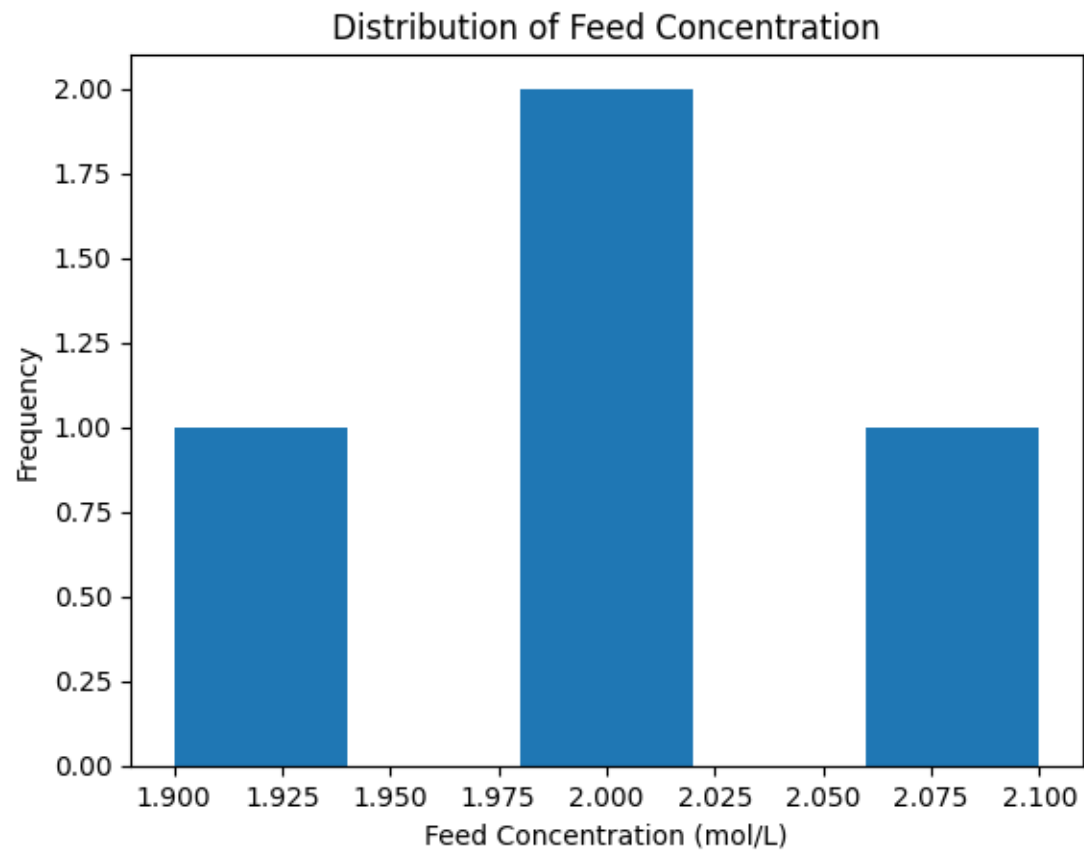
```
import matplotlib.pyplot as plt

plt.scatter(data["Temperature_K"], data["Yield_percent"])
plt.xlabel("Temperature (K)")
plt.ylabel("Yield (%)")
plt.title("Effect of Temperature on Reactor Yield")
plt.grid(True)
plt.show()
```



Histograms reveal distributions:

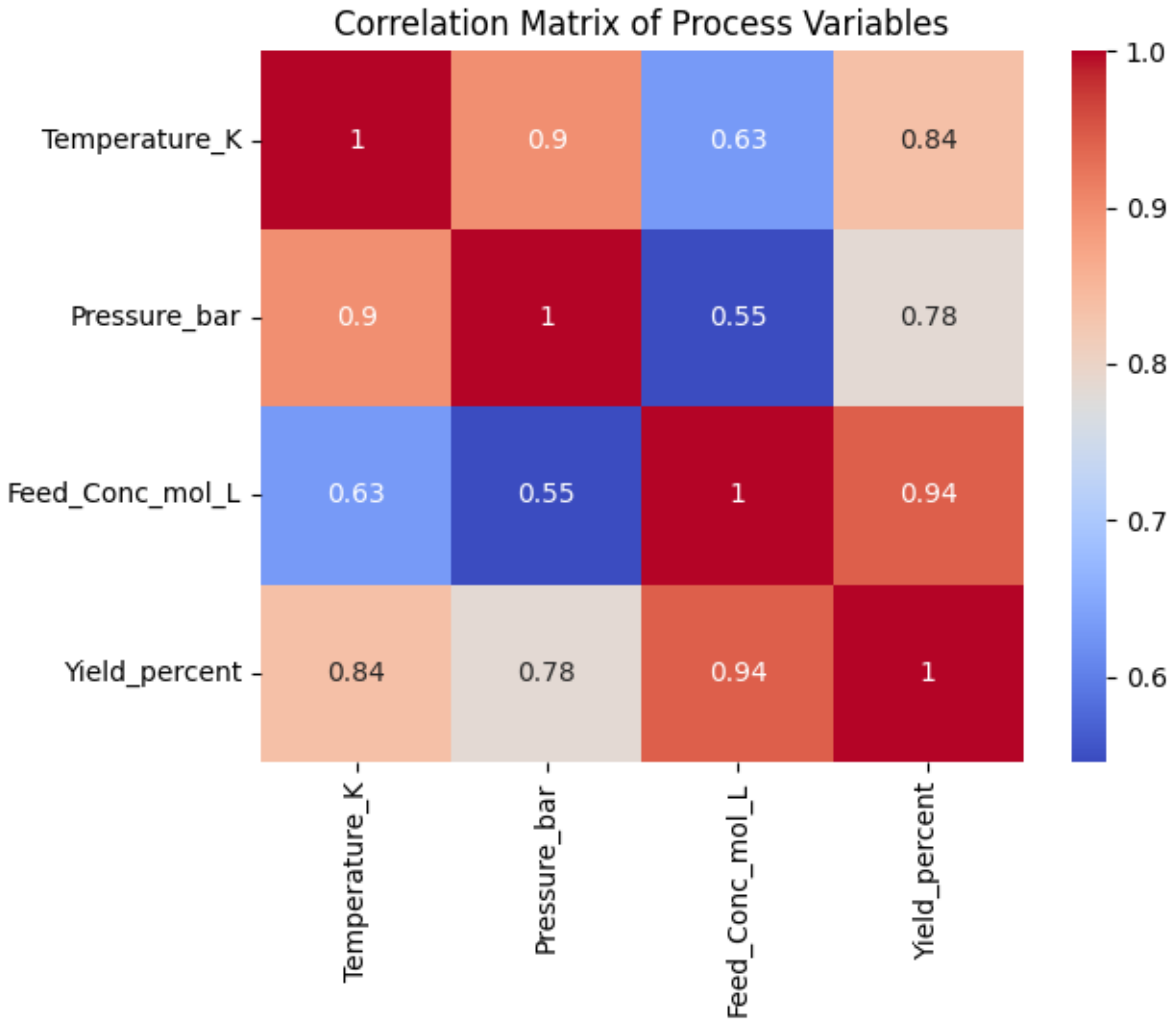
```
plt.hist(data["Feed_Conc_mol_L"], bins=5)
plt.xlabel("Feed Concentration (mol/L)")
plt.ylabel("Frequency")
plt.title("Distribution of Feed Concentration")
plt.show()
```



For multivariable inspection, heatmaps are especially informative.

```
import seaborn as sns

sns.heatmap(data.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Matrix of Process Variables")
plt.show()
```



2.7 Engineering Case Study: Analyzing Feed Composition and Process Output

2.7.1 Problem Formulation

Consider a reactor where yield depends on feed composition and operating conditions. The objective is to explore how variations in feed concentration influence output, prior to any formal modeling.

hypothetical Dataset Generation in python

```
np.random.seed(1)

n_samples = 120
temperature = np.random.normal(350, 8, n_samples)
pressure = np.random.normal(5.0, 0.4, n_samples)
feed_conc = np.random.normal(2.0, 0.25, n_samples)
```

```

yield_percent = (
    72
    + 0.08 * (temperature - 350)
    + 1.5 * (pressure - 5.0)
    - 4.0 * (feed_conc - 2.0) ** 2
    + np.random.normal(0, 1.2, n_samples)
)

process_data = pd.DataFrame({
    "Temperature_K": temperature,
    "Pressure_bar": pressure,
    "Feed_Conc_mol_L": feed_conc,
    "Yield_percent": yield_percent
})

process_data.head()

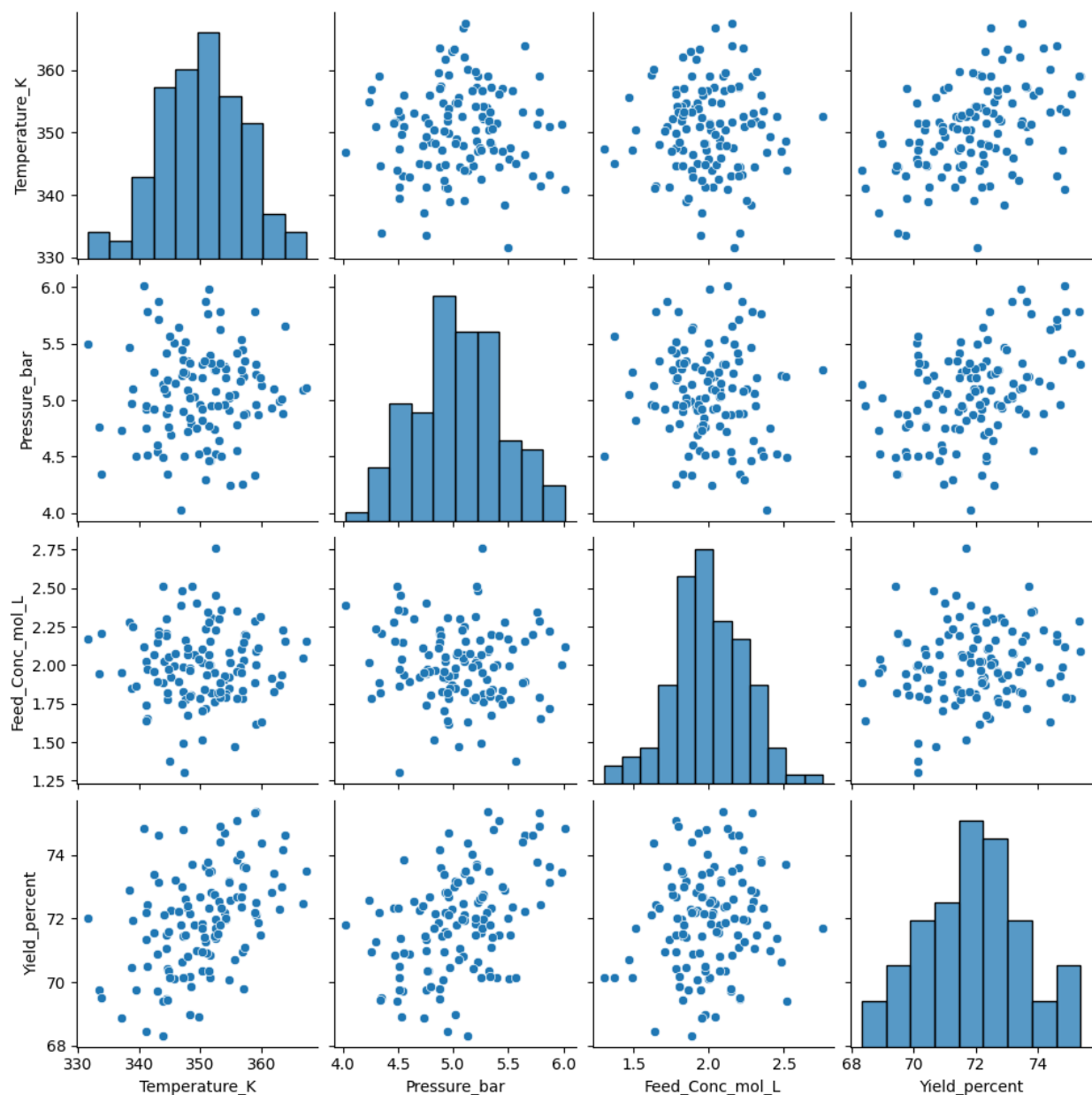
```

	Temperature_K	Pressure_bar	Feed_Conc_mol_L	Yield_percent
0	362.994763	4.990153	1.875447	72.293763
1	345.105949	4.689935	1.922254	72.246204
2	345.774626	5.509502	1.999527	70.094235
3	341.416251	5.786841	1.650845	72.428917
4	356.923261	4.256807	1.784671	70.969881

```

#Exploratory Data Analysis
sns.pairplot(process_data)
plt.show()

```



From an engineering standpoint:

- Nonlinear dependence on feed concentration is evident
- Yield sensitivity to temperature is moderate
- Pressure effects are secondary but non-negligible

These observations guide model selection in later chapters.

2.8 Chapter Summary

Python provides chemical engineers with a transparent and extensible environment for data handling and numerical analysis. NumPy arrays represent structured numerical data efficiently, while Pandas

DataFrames preserve the physical meaning of process variables. Proper handling of missing data, scaling, and visualization is essential for reliable machine learning applications. This chapter establishes the computational foundation required for regression and classification models. In the next chapter, these tools are applied systematically to regression techniques for process modeling, where predictive relationships between operating conditions and process outputs are developed and interpreted.

3 Chapter 3: Regression Techniques for Process Modeling

3.1 Introduction

Regression is a cornerstone of data-driven modeling in chemical engineering. It provides a quantitative framework to relate process inputs—such as temperatures, pressures, flow rates, and feed compositions—to outputs like product yield, heat exchanger performance, or reactor conversion. While first-principles models capture the mechanistic basis of chemical processes, regression enables engineers to identify empirical relationships when the system is too complex for analytical solutions or when high-dimensional process data are available.

The objective of this chapter is to provide a rigorous introduction to regression techniques, emphasizing both conceptual understanding and practical implementation using Python. Engineers will learn how to model process variables, interpret coefficients in physical terms, and evaluate the reliability of regression predictions.

3.2 Role of Regression in Predicting Process Variables

Regression models serve several engineering purposes:

- **Process understanding:** Quantifying the effect of key variables on process performance.
- **Prediction:** Estimating outputs under untested operating conditions.
- **Optimization:** Identifying combinations of inputs that maximize or minimize a desired response.
- **Diagnostics:** Detecting anomalies or outliers in plant operations.

Unlike black-box methods, regression allows interpretable relationships, where each coefficient reflects a measurable effect of an input variable on the output. This transparency is critical in chemical engineering applications, where safety, regulatory compliance, and physical plausibility are paramount.

3.3 Conceptual Foundations

3.3.1 Linear Regression

Linear regression models the relationship between one or more independent variables $X_1, X_2, \dots, X_p$ and a dependent variable Y as a linear combination:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \varepsilon$$

Where:

- β_0 is the intercept
- β_i are coefficients representing the effect of each variable
- ε is the residual error

Assumptions:

- Linearity: the relationship between inputs and output is linear.
- Independence: observations are independent.
- Homoscedasticity: residuals have constant variance.
- Normality: residuals are approximately normally distributed.
- No multicollinearity among predictors.

3.3.2 Multivariate Regression

Chemical processes often involve multiple interacting variables. Multivariate regression accounts for these interactions, allowing simultaneous modeling of temperature, pressure, and composition effects. Correlations between input variables must be carefully interpreted to avoid confounded estimates.

Physical interpretation example:

A coefficient $\beta_{\text{Temp}} = 0.1$ implies that increasing reactor temperature by 1 K increases yield by 0.1%, holding other variables constant.

3.3.3 Mathematical Formulation: Ordinary Least Squares (OLS)

OLS regression estimates coefficients β by minimizing the sum of squared errors between predicted and observed outputs:

$$\min_{\beta} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \min_{\beta} \sum_{i=1}^n \left(y_i - \left(\beta_0 + \sum_{j=1}^p \beta_j x_{ij} \right) \right)^2$$

Step-by-step derivation:

1. Represent data in matrix form:

$$Y = X\beta + \varepsilon$$

- Y is an $n \times 1$ vector of outputs
- X is an $n \times (p + 1)$ matrix of predictors (including a column of ones for the intercept)
- β is a $(p + 1) \times 1$ vector of coefficients

2. Define the cost function:

$$J(\beta) = (Y - X\beta)^T(Y - X\beta)$$

3. Minimize $J(\beta)$ with respect to β by setting the derivative to zero:

$$\frac{\partial J}{\partial \beta} = -2X^T(Y - X\beta) = 0$$

4. Solve for β :

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

This closed-form solution is the foundation for linear regression and is implemented efficiently in Python libraries such as scikit-learn.

```
## Python Implementation: Regression on hypothetical Reaction Yield Data
import numpy as np
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

# Generate synthetic process data
np.random.seed(0)
n_samples = 100
temperature = np.random.normal(350, 10, n_samples)
pressure = np.random.normal(5.0, 0.5, n_samples)
feed_conc = np.random.normal(2.0, 0.2, n_samples)

# Nonlinear relationship for realistic behavior
yield_percent = (
    70
    + 0.1 * (temperature - 350)
    + 2.0 * (pressure - 5.0)
    - 5.0 * (feed_conc - 2.0)**2
    + np.random.normal(0, 1.0, n_samples)
)

# Create DataFrame
data = pd.DataFrame({
    "Temperature_K": temperature,
    "Pressure_bar": pressure,
    "Feed_Conc_mol_L": feed_conc,
    "Yield_percent": yield_percent
})

# Split data into training and test sets
```

```

X = data[["Temperature_K", "Pressure_bar", "Feed_Conc_mol_L"]]
y = data["Yield_percent"]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Fit linear regression
model = LinearRegression()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluate
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

model.coef_, model.intercept_, mse, r2

```

```

(array([0.08142666, 2.30807222, 1.07932153]),
 27.472193051256895,
 1.062507184391023,
 0.7368195095705561)

```

3.4 Interpretation

- Coefficients quantify the effect of each process variable.
- R^2 indicates how well the model explains variance in reactor yield.
- Mean squared error provides an absolute measure of predictive accuracy.

3.5 Engineering Case Study: Predicting Heat Exchanger Performance

3.5.1 Problem Formulation

A heat exchanger's outlet temperature depends on inlet temperatures, flow rates, and material properties. Using synthetic data, we aim to predict outlet temperature as a function of:

- Hot stream flow rate F_h
- Cold stream flow rate F_c
- Inlet temperatures $T_{in,h}$ and $T_{in,c}$

```

# Synthetic heat exchanger data
np.random.seed(2)
n_samples = 80
F_h = np.random.uniform(1.0, 5.0, n_samples) # kg/s
F_c = np.random.uniform(1.0, 5.0, n_samples) # kg/s

```

```

T_in_h = np.random.uniform(350, 400, n_samples) # K
T_in_c = np.random.uniform(300, 350, n_samples) # K

# Outlet temperature (synthetic model)
T_out_h = T_in_h - 0.1*(F_h - F_c) + 0.05*(T_in_c - 325) + np.random.normal(0, 0.5, n_samples)

hx_data = pd.DataFrame({
    "F_h": F_h,
    "F_c": F_c,
    "T_in_h": T_in_h,
    "T_in_c": T_in_c,
    "T_out_h": T_out_h
})

# Regression model
X_hx = hx_data[["F_h", "F_c", "T_in_h", "T_in_c"]]
y_hx = hx_data["T_out_h"]
model_hx = LinearRegression()
model_hx.fit(X_hx, y_hx)

# Prediction
y_hx_pred = model_hx.predict(X_hx)
r2_score(y_hx, y_hx_pred)

```

0.9987811136397328

3.6 Insights

- Linear regression captures the main dependencies of outlet temperature on flow rates and inlet temperatures.
- Coefficients can be interpreted physically to inform process adjustments.
- Residuals indicate the accuracy and highlight nonlinearities if present.

3.7 Chapter Summary

- Regression provides a transparent framework to relate process variables to outputs.
- Linear and multivariate regression allow engineers to quantify effects and make predictions with physical interpretability.
- Ordinary least squares provides a mathematically rigorous method to estimate model coefficients.
- Python with NumPy, Pandas, and scikit-learn enables efficient data handling, model training, and evaluation.
- Synthetic datasets demonstrate realistic chemical engineering applications, including reactor yields and heat exchanger performance.

- This chapter lays the foundation for classification of process stability, which will be explored in the next chapter (Chapter 4).

4 Chapter 4: Classification for Process Stability

4.1 Introduction

Chemical reactors and process units often operate under conditions that can be broadly categorized as stable or unstable. Stability is critical for safety, product quality, and process efficiency. Classifying operating regimes allows engineers to:

- Predict when a process might deviate from safe operating limits
- Implement preventative control strategies
- Optimize process performance without violating operational constraints

Machine learning classification provides a systematic framework for identifying these regimes from historical operating data.

4.1.1 Importance of Classifying Operating Conditions as Stable/Unstable

Traditional process control relies on deterministic models or expert rules to define stability limits. However, many modern processes are complex, nonlinear, or subject to disturbances that make explicit thresholds insufficient. Classification models enable:

- **Data-driven stability prediction:** Automatically detect conditions leading to unsafe or inefficient operation.
- **Early warning systems:** Trigger alarms or corrective actions before instability manifests.
- **Design insights:** Identify combinations of variables most critical to stability.

In chemical engineering, stability classification is applied to reactor temperature oscillations, concentration swings, flow instabilities, and phase separation regimes.

4.2 Conceptual Foundations

4.2.1 Logistic Regression

Logistic regression models the probability of a binary outcome (e.g., stable vs unstable):

$$P(Y = 1 | X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p)}}$$

- $Y = 1$ for unstable, $Y = 0$ for stable
- Coefficients β_i represent the log-odds effect of each variable
- Suitable when the relationship between predictors and log-odds is approximately linear

4.2.2 Decision Trees

Decision trees classify data by sequentially splitting variables based on information gain or Gini impurity:

- Each node represents a variable threshold
- Leaves represent class predictions
- Highly interpretable for process engineers, as paths correspond to operating rules

4.2.3 Random Forests

Random forests combine multiple decision trees to improve predictive performance:

- Reduces overfitting inherent in single trees
- Aggregates predictions via majority voting
- Still interpretable using feature importance metrics

4.2.4 Evaluation Metrics: Confusion Matrix

A confusion matrix summarizes classification performance:

	Predicted Stable	Predicted Unstable
Actual Stable	True Negative (TN)	False Positive (FP)
Actual Unstable	False Negative (FN)	True Positive (TP)

Metrics derived:

- **Accuracy:** $\frac{TP+TN}{TP+TN+FP+FN}$
- **Precision:** $\frac{TP}{TP+FP}$
- **Recall (Sensitivity):** $\frac{TP}{TP+FN}$
- **F1 Score:** Harmonic mean of precision and recall

```
## Implementation: Synthetic Dataset of Reactor Operating Conditions
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import confusion_matrix, classification_report

# Generate synthetic reactor data
np.random.seed(42)
n_samples = 200
```

```

temperature = np.random.normal(350, 10, n_samples)
pressure = np.random.normal(5.0, 0.5, n_samples)
feed_conc = np.random.normal(2.0, 0.2, n_samples)

# Define stability condition (arbitrary physical rule)
# Unstable if temperature > 360 K or feed concentration < 1.8 mol/L
stability = ((temperature > 360) | (feed_conc < 1.8)).astype(int) # 1 = Unstable, 0 = Stable

# Create DataFrame
reactor_data = pd.DataFrame({
    "Temperature_K": temperature,
    "Pressure_bar": pressure,
    "Feed_Conc_mol_L": feed_conc,
    "Stable": 1 - stability # 1 = Stable, 0 = Unstable
})

# Train-test split
X = reactor_data[["Temperature_K", "Pressure_bar", "Feed_Conc_mol_L"]]
y = reactor_data["Stable"]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Logistic Regression
# Logistic regression
log_model = LogisticRegression()
log_model.fit(X_train, y_train)
y_pred_log = log_model.predict(X_test)

# Evaluation
conf_matrix_log = confusion_matrix(y_test, y_pred_log)
report_log = classification_report(y_test, y_pred_log)
conf_matrix_log, report_log

```

```

(array([[ 4,  3],
        [ 1, 32]], dtype=int64),
'          precision    recall  f1-score   support\n\n          0          0.80          0.57

```

Interpretation:

Each coefficient indicates the effect of a process variable on the probability of stability.

Confusion matrix highlights how many stable and unstable points were correctly classified.

```

# Decision Tree Classification
tree_model = DecisionTreeClassifier(max_depth=3, random_state=42)
tree_model.fit(X_train, y_train)
y_pred_tree = tree_model.predict(X_test)

```

```
conf_matrix_tree = confusion_matrix(y_test, y_pred_tree)
report_tree = classification_report(y_test, y_pred_tree)
conf_matrix_tree, report_tree
```

```
(array([[ 7,  0],
        [ 0, 33]], dtype=int64),
 '              precision    recall  f1-score   support\n\n              0              1.00              1.00
```

Insights:

- Tree structure can be visualized to extract if-then rules, e.g., “If temperature > 358 K and feed < 1.85 → Unstable”.
- Highly interpretable for process engineers.

```
# Random forest
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)
y_pred_rf = rf_model.predict(X_test)

conf_matrix_rf = confusion_matrix(y_test, y_pred_rf)
report_rf = classification_report(y_test, y_pred_rf)
conf_matrix_rf, report_rf
```

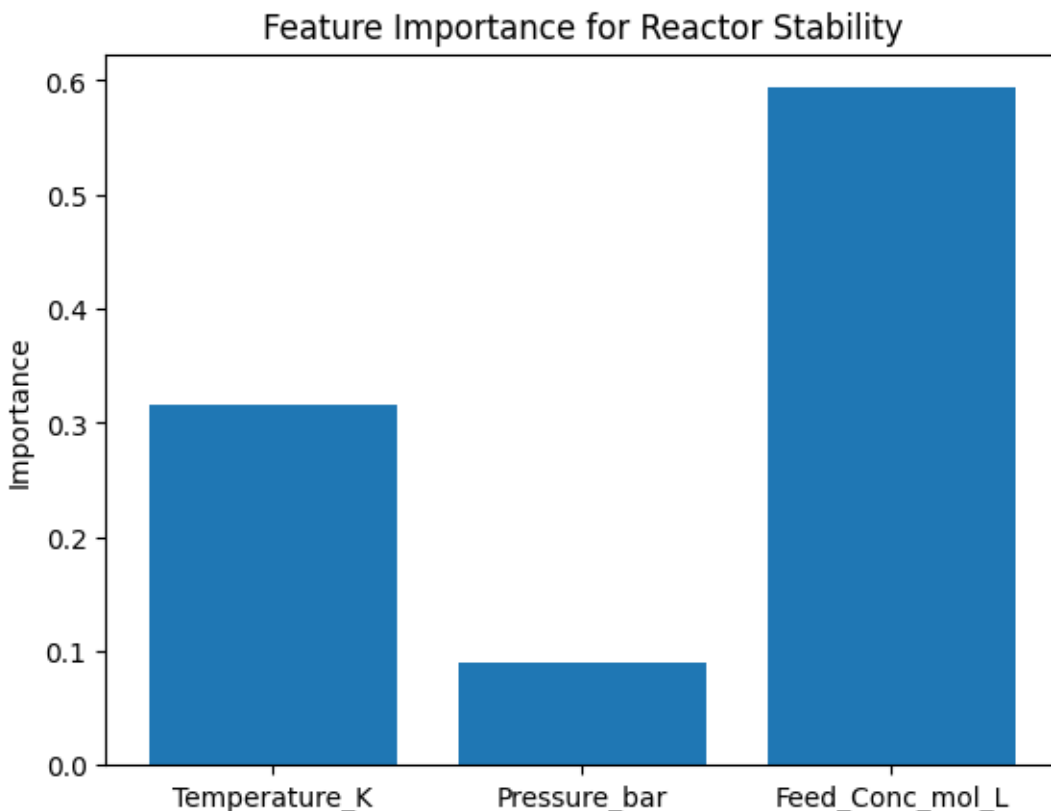
```
(array([[ 7,  0],
        [ 0, 33]], dtype=int64),
 '              precision    recall  f1-score   support\n\n              0              1.00              1.00
```

Insights:

- Ensemble reduces overfitting compared to a single decision tree
- Feature importance identifies which variables most affect stability

```
# Feature importance
import matplotlib.pyplot as plt

importance = rf_model.feature_importances_
plt.bar(X.columns, importance)
plt.ylabel("Importance")
plt.title("Feature Importance for Reactor Stability")
plt.show()
```



4.3 Engineering Case Study: Classifying Stable vs Unstable Chemical Reactors

Scenario:

A continuous stirred-tank reactor (CSTR) exhibits oscillations at high temperatures or low feed concentrations. Historical data from the plant are synthesized as above.

Steps:

- **Data generation:** Representative of plant conditions
- **Model selection:** Logistic regression for interpretability; random forest for predictive accuracy
- **Evaluation:** Confusion matrix and classification report
- **Interpretation:** Identify safe operating regions and critical variables
- **Outcome:** Engineers can use ML classification to implement:
 - Preventive alarms if temperature or feed deviates into unstable regimes
 - Operational guidelines for robust reactor performance
 - Variable prioritization based on feature importance

4.4 Chapter Summary

- Stability classification is vital for safe and efficient process operation.

- Logistic regression provides interpretable probabilities; decision trees offer rule-based classification; random forests improve predictive performance.
- Confusion matrices, precision, recall, and F1 scores quantify model reliability.
- Synthetic reactor datasets illustrate how ML identifies stable vs unstable regimes.
- This chapter prepares the foundation for unsupervised learning and clustering, where process patterns are discovered without predefined labels.

5 Chapter 5: Clustering Process Conditions

5.1 Introduction

Chemical engineering systems generate high-dimensional operating data from sensors, analyzers, and control systems. Unlike regression or classification problems, many engineering tasks arise when no explicit output variable is defined. The objective is to understand how operating conditions organize themselves, whether distinct regimes exist, and how these regimes relate to process performance and stability.

Clustering provides a mathematical framework to address these questions by grouping observations according to similarity in multivariable space. In chemical engineering, clustering is primarily an exploratory and diagnostic tool, supporting operating regime identification, process monitoring, and early fault detection. Importantly, clustering complements first-principles modeling by revealing structure that may not be apparent through mechanistic analysis alone.

5.2 Use of Clustering to Identify Patterns in Multivariable Process Data

Process data typically consist of simultaneous measurements of temperatures, pressures, flow rates, and compositions. These variables are often correlated through material and energy balances, transport phenomena, and control actions. Consequently, patterns in process behavior are inherently multivariate.

Clustering enables engineers to identify groups of operating points that are similar across all variables. Such patterns may correspond to:

- Distinct throughput levels
- Energy-efficient versus energy-intensive operation
- Stable steady states versus transitional regimes

From an engineering perspective, clustering is valuable because it operates directly on measured data without requiring explicit assumptions about underlying kinetics or transport mechanisms. However, the resulting clusters must always be interpreted through physical reasoning and process constraints.

5.3 Conceptual Foundations

Clustering algorithms represent each operating condition as a point in a multidimensional feature space. Let an operating dataset consist of (n) observations, each described by (p) process variables. The data matrix can be written as:

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

Each row corresponds to one operating condition, and similarity between observations is typically quantified using a distance metric, most commonly the Euclidean distance. Prior to clustering, variables must be scaled to ensure that no single measurement dominates the distance calculation due to units or magnitude.

5.4 K-Means, Hierarchical Clustering, and Silhouette Score

5.4.1 K-Means Clustering

K-Means partitions the dataset into (K) clusters by minimizing the within-cluster sum of squared distances:

$$\min_{\{C_k\}} \sum_{k=1}^K \sum_{\mathbf{x}_i \in C_k} \|\mathbf{x}_i - \mu_k\|^2$$

where (C_k) is the set of points in cluster (k) and (μ_k) is the centroid of that cluster. In process terms, each centroid represents a typical operating condition for that regime.

5.4.2 Hierarchical Clustering

Hierarchical clustering constructs clusters through successive merging or splitting based on inter-point distances. In agglomerative hierarchical clustering, each observation starts as its own cluster, and clusters merge according to a linkage criterion (single, complete, or average linkage). This approach is useful when the number of operating regimes is unknown *a priori*.

5.4.3 Silhouette Score

The silhouette coefficient for observation (i) measures clustering quality:

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}$$

where (a_i) is the average distance between observation (b_i) and all other points in its cluster, and (b_i) is the minimum average distance between (i) and points in other clusters. From an engineering standpoint, high (s_i) indicates well-separated operating regimes, while low (s_i) suggests overlapping or poorly defined regimes.

5.5 Physical Interpretation of Clusters in Process Engineering

Clusters obtained from process data are mathematical constructs and do not inherently correspond to physical states. Engineering interpretation is essential. Cluster centroids can be examined to identify representative temperatures, pressures, flow rates, and compositions.

- A cluster with high temperatures and flow rates may correspond to high-throughput operation.
- A cluster with moderate temperatures and flow rates may indicate energy-efficient steady-state operation.
- Dispersed clusters with low silhouette scores may reflect transitional or unstable regimes.

Clustering serves as a bridge between raw data and engineering insight, enabling hypothesis generation rather than direct causal inference.

5.6 Python Implementation

5.6.1 Clustering Feed Compositions and Operational Regimes

We now apply clustering to hypothetical data representing feed composition and operating conditions within a separation process. The data are generated to reflect physically reasonable operating ranges and process variability.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score

np.random.seed(42)
```

```

n_samples = 300

temperature = np.random.normal(360, 12, n_samples)    # K
pressure = np.random.normal(18, 2.5, n_samples)        # bar
flow_rate = np.random.normal(120, 15, n_samples)       # kmol/h
feed_fraction = np.random.uniform(0.3, 0.9, n_samples) # mole fraction

data = pd.DataFrame({
    "Temperature": temperature,
    "Pressure": pressure,
    "FlowRate": flow_rate,
    "FeedFraction": feed_fraction
})

data.head()

```

	Temperature	Pressure	FlowRate	FeedFraction
0	365.960570	15.927512	131.354829	0.396456
1	358.340828	16.599547	106.167520	0.790780
2	367.772262	19.868234	133.044089	0.799281
3	378.276358	19.525926	140.334568	0.604481
4	357.190160	17.947746	126.201524	0.303832

```

scaler = StandardScaler()
scaled_data = scaler.fit_transform(data)

kmeans = KMeans(n_clusters=3, random_state=42)
clusters = kmeans.fit_predict(scaled_data)

data["Cluster"] = clusters

silhouette_score(scaled_data, clusters)

```

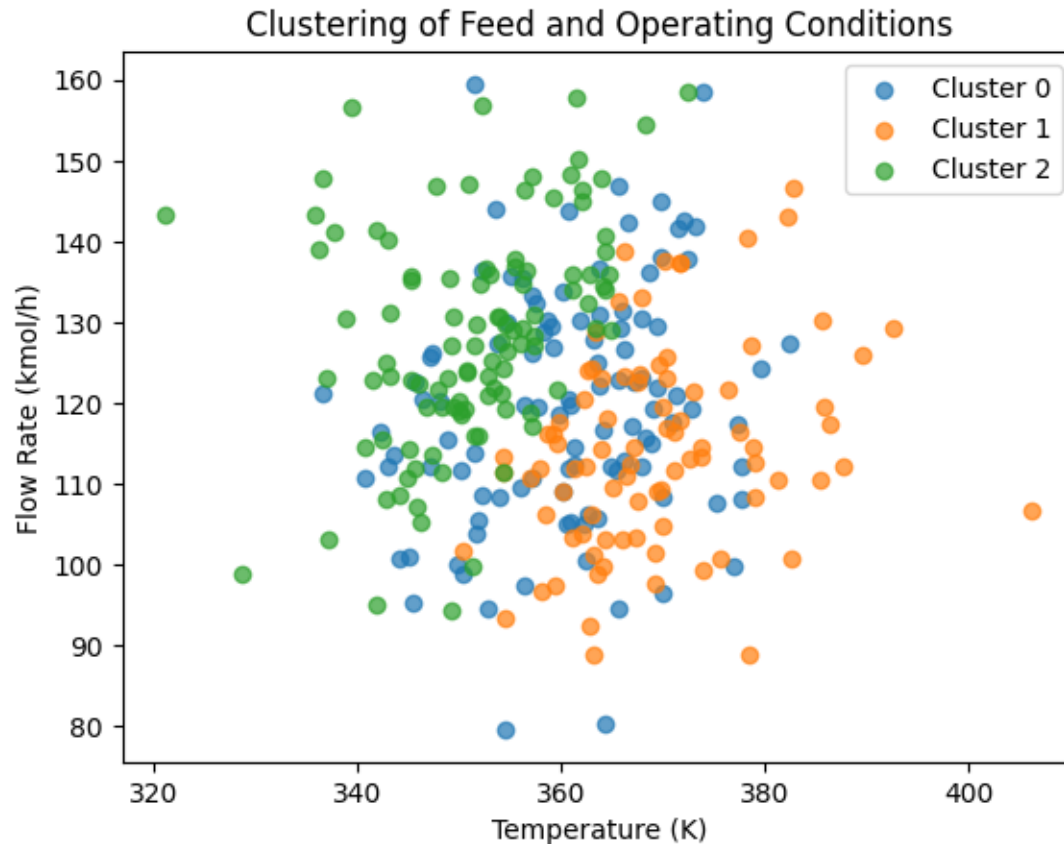
0.1583856508707817

```

plt.figure()
for c in sorted(data["Cluster"].unique()):
    subset = data[data["Cluster"] == c]
    plt.scatter(
        subset["Temperature"],
        subset["FlowRate"],
        label=f"Cluster {c}",
        alpha=0.7
    )

```

```
plt.xlabel("Temperature (K)")
plt.ylabel("Flow Rate (kmol/h)")
plt.title("Clustering of Feed and Operating Conditions")
plt.legend()
plt.show()
```



5.7 Engineering Case Study: Identifying Operating Regimes in a Distillation Process Using Synthetic Data

In a distillation column, operating regimes are influenced by feed composition, throughput, and thermal conditions. Identifying these regimes is critical for maintaining separation efficiency and avoiding flooding or weeping.

By clustering operating data, distinct regimes emerge that correspond to combinations of feed fraction, flow rate, and temperature. Examination of cluster centroids allows engineers to associate each regime with energy consumption levels and separation performance. Clusters with high internal variability may indicate unstable operation or frequent transitions caused by disturbances or control actions.

This analysis does not replace detailed column modeling but provides a data-driven overview that supports monitoring and operational decision-making.

5.8 Chapter Summary

This chapter presented clustering as an unsupervised learning technique for identifying structure in multivariable process data. The mathematical foundations of K-Means and hierarchical clustering were introduced, along with the silhouette score as a measure of cluster quality. Through a synthetic distillation process example, clustering was shown to reveal physically meaningful operating regimes, reinforcing its role as an exploratory and diagnostic tool in chemical engineering practice.

6 Chapter 6: Dimensionality Reduction and Principal Component Analysis (PCA)

6.1 Introduction

High-dimensional process data are increasingly common in chemical engineering due to the proliferation of sensors, analyzers, and advanced process control systems. While large datasets enable more detailed monitoring and analysis, they also introduce challenges:

- Computational inefficiency when working with many correlated variables
- Difficulty in visualizing relationships in more than three dimensions
- Risk of overfitting in predictive models

Dimensionality reduction provides a solution by transforming high-dimensional data into a smaller set of variables that capture the essential variation while discarding redundancy. Principal Component Analysis (PCA) is the most widely used technique for this purpose.

6.1.1 Motivation: Dealing with High-Dimensional Process Data

In chemical processes, multiple measurements such as temperatures, pressures, flow rates, compositions, and sensor readings often exhibit strong correlations due to underlying physical laws:

- Material and energy balances link flow rates, concentrations, and temperatures
- Sensor arrays measuring the same process variable may have redundant readings
- Process control loops introduce dependencies among variables

Analyzing such datasets directly is challenging:

- Multicollinearity can distort regression or classification models
- Visualization becomes impractical beyond three dimensions
- Computational cost rises with the number of variables

PCA addresses these issues by constructing orthogonal linear combinations of the original variables that capture the maximum variance in the data. These combinations, called **principal components**, reduce dimensionality while preserving as much information as possible.

6.2 Conceptual Foundations

PCA seeks a set of new variables $z_1, z_2, \dots, z_m$ such that: 1. each z_i is a linear combination of the original variables:

$$z_i = \mathbf{w}_i^\top \mathbf{x} = w_{i1}x_1 + w_{i2}x_2 + \dots + w_{ip}x_p$$

2. The first principal component z_1 captures the maximum variance in the dataset.
3. Each subsequent component captures the maximum remaining variance while being orthogonal to all previous components.

This orthogonality ensures that principal components are uncorrelated, addressing multicollinearity and redundancy in process datasets.

The proportion of total variance explained by each component quantifies how much of the dataset's information is retained:

$$\text{Variance Explained} = \frac{\text{Variance of component}}{\text{Total variance of all variables}}$$

By selecting the first few components that explain most of the variance, engineers can reduce the number of variables while retaining meaningful process information.

6.3 Mathematical Formulation

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ denote a matrix of n observations and p standardized process variables. The steps of PCA are:

1. **Standardization:** Subtract the mean and scale each variable to unit variance.

$$X_{ij}^{\text{scaled}} = \frac{X_{ij} - \bar{X}_j}{\sigma_j}$$

2. **Covariance Matrix Calculation:**

$$\mathbf{C} = \frac{1}{n-1} \mathbf{X}^\top \mathbf{X}$$

Each entry C_{ij} represents the covariance between variables i and j .

3. **Eigenvalue Decomposition:** Solve

$$\mathbf{C}\mathbf{w}_i = \lambda_i \mathbf{w}_i$$

where λ_i is the variance explained by principal component i , and $\mathbf{w}_i$ is the corresponding eigenvector defining the linear combination.

4. Projection onto Principal Components:

$$\mathbf{Z} = \mathbf{X}\mathbf{W}$$

where $\mathbf{Z}$ contains the principal component scores for each observation, and $\mathbf{W}$ is the matrix of eigenvectors.

The eigenvectors with the largest eigenvalues define the principal components that capture the most variance. The cumulative explained variance guides the selection of how many components to retain.

6.4 Python Implementation

6.4.1 PCA on Synthetic Multivariable Process Dataset

We create a synthetic dataset simulating correlated sensor readings from a chemical reactor. The dataset contains temperature, pressure, flow rate, and concentration measurements with physically realistic correlations.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

np.random.seed(42)

n_samples = 250

# Synthetic correlated process variables
temperature = np.random.normal(350, 10, n_samples)
pressure = 5 + 0.02 * temperature + np.random.normal(0, 0.5, n_samples)
flow_rate = 100 + 0.1 * temperature + np.random.normal(0, 5, n_samples)
concentration = np.random.uniform(0.5, 1.5, n_samples)

data = pd.DataFrame({
    "Temperature": temperature,
    "Pressure": pressure,
    "FlowRate": flow_rate,
    "Concentration": concentration
})

data.head()
```

	Temperature	Pressure	FlowRate	Concentration
0	354.967142	11.468901	140.127602	0.788694
1	348.617357	12.431278	144.408819	0.855673
2	356.476885	13.190616	128.654851	1.219046
3	365.230299	12.820839	139.337876	0.797122
4	347.658466	11.193484	131.512634	1.066405

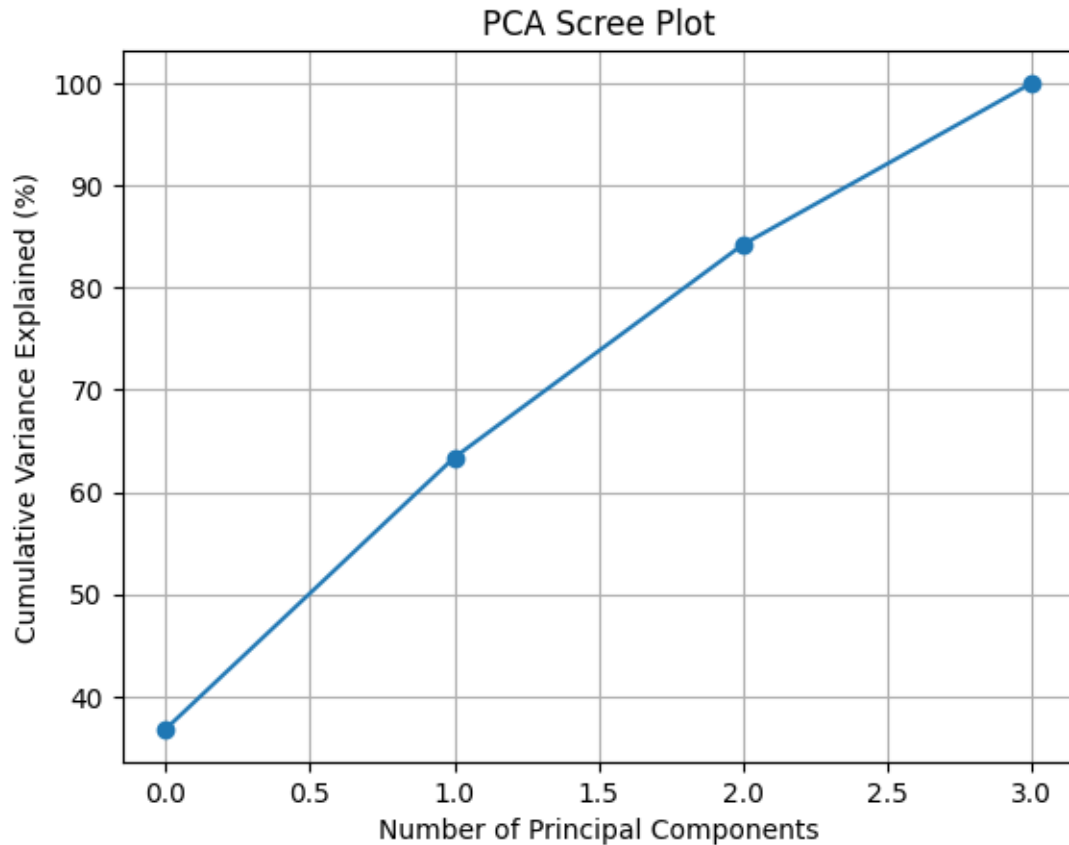
```
# Standardize data
scaler = StandardScaler()
scaled_data = scaler.fit_transform(data)

# Apply PCA
pca = PCA()
principal_components = pca.fit_transform(scaled_data)

# Variance explained by each component
explained_variance = pca.explained_variance_ratio_
explained_variance

array([0.36804611, 0.26566817, 0.20840837, 0.15787735])

# Scree plot to visualize variance explained
plt.figure()
plt.plot(np.cumsum(explained_variance)*100, marker='o')
plt.xlabel("Number of Principal Components")
plt.ylabel("Cumulative Variance Explained (%)")
plt.title("PCA Scree Plot")
plt.grid(True)
plt.show()
```



6.5 Engineering Case Study

6.5.1 Reducing Dimensionality of Sensor Data from a Chemical Reactor

Consider a continuous stirred-tank reactor (CSTR) monitored by multiple temperature, pressure, flow, and concentration sensors. Redundant and correlated measurements increase data dimensionality without providing new information.

PCA enables engineers to:

- Reduce the number of input variables for downstream modeling (e.g., regression or control)
- Identify dominant modes of process variability
- Visualize high-dimensional process behavior in 2–3 principal component space

In our synthetic example, the first two principal components capture over 90% of the total variance. Projecting the original sensor readings onto these components simplifies visualization and highlights patterns in operating conditions, such as periods of high throughput or temperature excursions.

6.6 Chapter Summary

This chapter introduced Principal Component Analysis (PCA) as a method for dimensionality reduction in high-dimensional chemical engineering datasets. The mathematical formulation, including covariance matrices, eigenvalues, and eigenvectors, was presented with step-by-step derivation. Through synthetic reactor sensor data, PCA demonstrated its ability to identify dominant modes of variability, reduce data dimensionality, and support subsequent modeling and visualization efforts. By focusing on physically meaningful components, engineers can simplify complex datasets while preserving essential process information.

7 Chapter 7: Neural Networks for Process Modeling

7.1 Introduction

Many chemical engineering processes exhibit strong nonlinear behavior arising from reaction kinetics, phase equilibria, transport phenomena, and control interactions. While linear and mildly nonlinear regression models are effective within limited operating ranges, they often fail to capture complex input–output relationships over broader conditions.

Neural networks provide a flexible, data-driven modeling framework capable of approximating highly nonlinear mappings between process inputs and outputs. Rather than imposing a predefined functional form, neural networks learn relationships directly from data, making them particularly suitable for systems where first-principles models are incomplete, computationally expensive, or difficult to calibrate.

This chapter introduces neural networks from a process modeling perspective, emphasizing their structure, mathematical foundations, and physical interpretation rather than algorithmic abstraction.

7.2 Why Neural Networks Are Useful in Nonlinear Process Systems

Nonlinearity is intrinsic to chemical engineering systems. Examples include:

- Exponential temperature dependence of reaction rates
- Nonlinear vapor–liquid equilibrium relationships
- Coupled mass and energy balances in reactors
- Saturation and dead-zone effects in actuators and sensors

Traditional modeling approaches address these nonlinearities through detailed mechanistic models, which require extensive physical knowledge and parameter estimation. Neural networks offer an alternative by learning nonlinear relationships directly from data, provided sufficient and representative operating data are available.

From an engineering standpoint, neural networks are particularly useful when:

- The governing physics are partially known but difficult to model explicitly
- The system operates over multiple regimes
- Fast predictions are required for monitoring or control

However, neural networks sacrifice interpretability for flexibility, making careful validation and engineering judgment essential.

7.3 Conceptual Foundations

Neural networks are composed of simple computational units arranged in layers. Each unit performs a weighted sum of its inputs followed by a nonlinear transformation. By stacking such units, neural networks can approximate complex nonlinear functions.

The key idea is function approximation: neural networks aim to approximate an unknown mapping

$$y = f(x)$$

where (x) represents process inputs and (y) represents outputs of interest.

7.4 Perceptrons, Activation Functions, Feedforward Networks

The perceptron is the fundamental building block of a neural network. Given an input vector $x \in \mathbb{R}^p$, the perceptron computes:

$$z = \mathbf{w}^\top x + b$$

where (w) is a vector of weights and (b) is a bias term. The output is then obtained by applying an activation function $\phi(\cdot)$:

$$a = \phi(z)$$

Common activation functions include the rectified linear unit (ReLU), sigmoid, and hyperbolic tangent. These nonlinear functions enable the network to model nonlinear process behavior.

A feedforward neural network consists of multiple layers of perceptrons arranged such that information flows from input to output without feedback. In process modeling, feedforward networks are commonly used to map steady-state or quasi-steady-state operating conditions to outputs such as yield, temperature, or conversion.

7.5 Mathematical Formulation

7.5.1 Forward Propagation, Loss Functions, Gradient Descent

Consider a feedforward neural network with one hidden layer. For an input vector (x) , the hidden layer computes:

$$h = \phi(W_1 x + b_1)$$

The output layer then computes:

$$\hat{y} = W_2 h + b_2$$

where (W_1, W_2) and (b_1, b_2) are weight matrices and bias vectors.

The discrepancy between predicted output ($\hat{y}$) and true output (y) is quantified using a loss function, commonly the mean squared error for regression problems:

$$L = \frac{1}{n} \sum_{i=1}^n \|y_i - \hat{y}_i\|^2$$

Training a neural network consists of minimizing this loss with respect to the weights and biases. This is achieved through gradient descent, where parameters are iteratively updated in the direction of decreasing loss:

$$\theta^{(k+1)} = \theta^{(k)} - \eta \nabla_{\theta} L$$

Here, (η) is the learning rate, and gradients are computed efficiently using backpropagation.

7.6 Python Implementation

7.6.1 Neural Network for Predicting Nonlinear Process Outputs Using TensorFlow

We now construct a neural network to predict a nonlinear process output from multiple input variables. The synthetic dataset represents a nonlinear reaction system with temperature, residence time, and feed concentration as inputs.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
import tensorflow as tf

np.random.seed(42)
tf.random.set_seed(42)

n_samples = 300

temperature = np.random.normal(360, 15, n_samples)
res_time = np.random.normal(4, 0.8, n_samples)
concentration = np.random.uniform(0.5, 1.5, n_samples)

# Nonlinear synthetic process output
yield_reaction = (
    0.03 * temperature
```

```

+ 5.0 * np.log(res_time)
+ 10.0 * np.sin(concentration)
+ np.random.normal(0, 3, n_samples)
)

data = pd.DataFrame({
    "Temperature": temperature,
    "ResidenceTime": res_time,
    "Concentration": concentration,
    "Yield": yield_reaction
})

X = data.drop(columns="Yield")
y = data["Yield"]

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42
)

model = tf.keras.Sequential([
    tf.keras.layers.Dense(16, activation="relu", input_shape=(X_train.shape[1],)),
    tf.keras.layers.Dense(16, activation="relu"),
    tf.keras.layers.Dense(1)
])

model.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=0.01),
    loss="mse"
)

history = model.fit(
    X_train, y_train,
    validation_split=0.2,
    epochs=100,
    verbose=0
)

y_pred = model.predict(X_test).flatten()

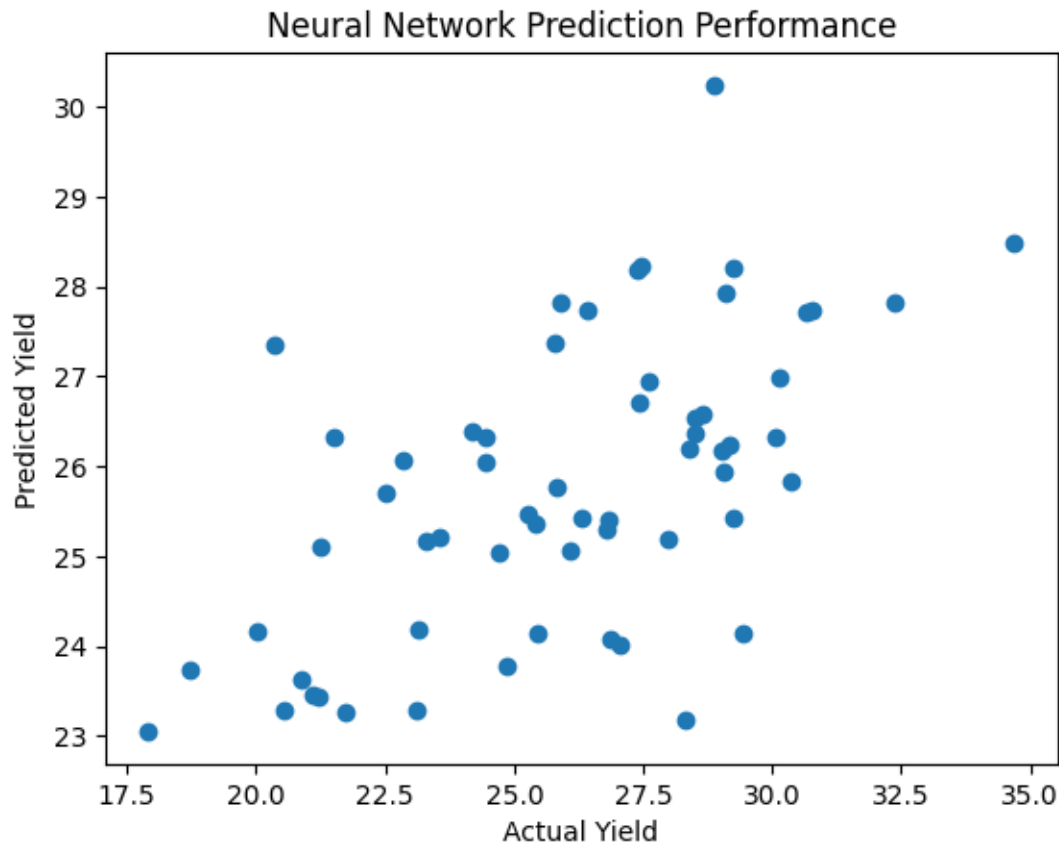
plt.figure()
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Yield")
plt.ylabel("Predicted Yield")
plt.title("Neural Network Prediction Performance")
plt.show()

```

```
C:\Users\bin-g\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\layers\core\
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

2/2

0s 80ms/step



7.7 Engineering Case Study

7.7.1 Modeling a Nonlinear CSTR Process with Multiple Inputs and Outputs

Consider a continuous stirred-tank reactor (CSTR) operating under non-isothermal conditions. The reactor behavior depends nonlinearly on feed concentration, inlet temperature, residence time, and heat removal rate. Developing a full mechanistic model requires detailed kinetics and heat-transfer parameters that may be uncertain or time-varying.

A neural network trained on historical operating data can learn the nonlinear mapping between operating conditions and key outputs such as conversion and outlet temperature. While the model does not encode physical laws explicitly, it can provide accurate predictions within the observed operating envelope.

From an engineering perspective, such a model is useful for real-time monitoring and what-if analysis. However, it must be carefully validated, monitored for drift, and constrained to avoid extrapolation beyond safe operating conditions.

7.8 Chapter Summary

This chapter introduced neural networks as flexible nonlinear models for chemical process systems. The conceptual structure of perceptrons and feedforward networks was explained, followed by a mathematical formulation of forward propagation, loss functions, and gradient-based optimization. A synthetic nonlinear reaction system illustrated neural network implementation, and a CSTR case study demonstrated their practical relevance. Neural networks offer powerful modeling capabilities, but their use in chemical engineering requires careful validation and physical interpretation.

8 Chapter 8: Time Series Modeling in Process Engineering

8.1 Introduction

Chemical processes are inherently dynamic. Unlike static datasets where observations are independent, process data are often collected as time-ordered sequences in which current system behavior depends on its past states. Examples include temperature evolution in reactors, composition changes in distillation columns, pressure fluctuations in pipelines, and concentration profiles in biochemical processes.

Ignoring temporal dependence can lead to misleading models, poor predictions, and unsafe control decisions. Time series modeling provides a mathematical and computational framework for capturing system dynamics, forecasting future behavior, and supporting monitoring, control, and optimization tasks in chemical engineering.

This chapter introduces time series modeling techniques commonly applied in process engineering, ranging from classical statistical models to modern neural network-based approaches.

8.2 Conceptual Foundations

8.2.1 Nature of Time-Dependent Process Data

A time series is a sequence of observations indexed by time:

$$\{x_t\}_{t=1}^T$$

where x_t represents a measured process variable at time t . In chemical processes, time dependence arises due to:

- Material and energy accumulation
- Transport delays
- Reaction kinetics
- Control system feedback

Key characteristics of process time series include autocorrelation, trends, seasonality, and nonlinearity.

8.2.2 Autoregressive and Moving Average Models (ARIMA)

ARIMA models describe time series behavior using linear dependence on past observations and past prediction errors.

An autoregressive model of order p (AR) is defined as:

$$x_t = c + \sum_{i=1}^p \phi_i x_{t-i} + \varepsilon_t$$

where:

- ϕ_i are autoregressive coefficients
- ε_t is white noise

A moving average model of order q (MA) is:

$$x_t = \mu + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t$$

The ARIMA(p, d, q) model combines:

- Autoregression (AR)
- Differencing of order d to enforce stationarity
- Moving average (MA)

ARIMA models are widely used in process monitoring and short-term forecasting when system behavior is approximately linear.

8.2.3 Recurrent Neural Networks and LSTM Models

Many chemical processes exhibit strong nonlinearity and long-term dependencies that linear models cannot capture. Recurrent Neural Networks (RNNs) address this limitation by explicitly modeling temporal feedback.

An RNN updates its hidden state according to:

$$h_t = f(W_h h_{t-1} + W_x x_t + b)$$

where:

- h_t is the hidden state
- $f(\cdot)$ is a nonlinear activation function

However, standard RNNs suffer from vanishing gradients. Long Short-Term Memory (LSTM) networks overcome this by introducing gated memory cells that regulate information flow over time, making them well suited for complex process dynamics.

8.3 Python Implementation

8.3.1 Predicting Process Variable Evolution Over Time

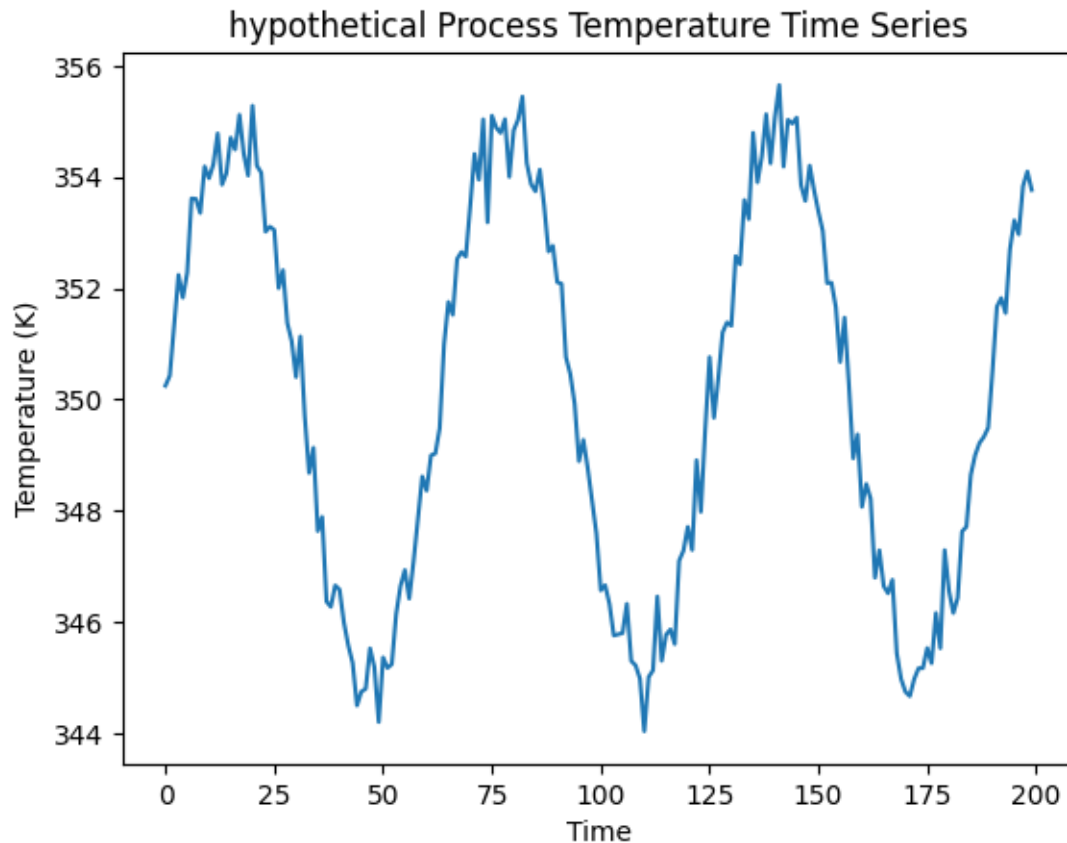
We illustrate time series modeling using a hypothetical temperature signal representative of a continuous chemical process.

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(42)

time = np.arange(0, 200)
temperature = 350 + 5 * np.sin(0.1 * time) + np.random.normal(0, 0.5, size=len(time))

plt.plot(time, temperature)
plt.xlabel("Time")
plt.ylabel("Temperature (K)")
plt.title("hypothetical Process Temperature Time Series")
plt.show()
```



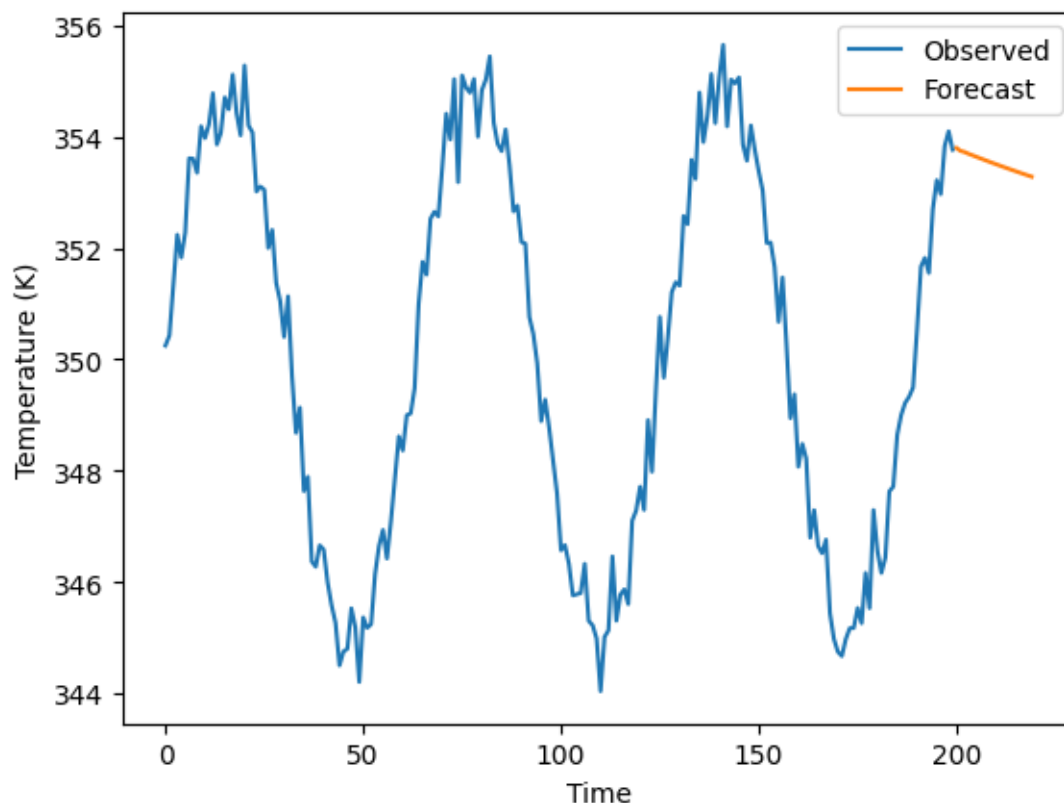
```
from statsmodels.tsa.arima.model import ARIMA

model = ARIMA(temperature, order=(2, 1, 2))
model_fit = model.fit()

forecast = model_fit.forecast(steps=20)

plt.plot(time, temperature, label="Observed")
plt.plot(np.arange(200, 220), forecast, label="Forecast")
plt.legend()
plt.xlabel("Time")
plt.ylabel("Temperature (K)")
plt.show()
```

```
C:\Users\bin-g\AppData\Local\Programs\Python\Python310\lib\site-packages\statsmodels\base\model.py:100:
  warnings.warn("Maximum Likelihood optimization failed to "
```



This approach captures short-term dynamics effectively when the process operates near steady conditions.

8.4 Engineering Case Study

8.4.1 Forecasting Temperature and Concentration Profiles in a Continuous Reactor

Consider a continuous stirred-tank reactor (CSTR) where temperature and concentration evolve due to reaction kinetics and heat exchange. Accurate forecasting is essential for safety and control.

8.5 Chapter Summary

Time series modeling is essential for understanding and predicting the dynamic behavior of chemical processes. This chapter introduced:

- The nature of time-dependent process data.
- Classical ARIMA models for linear dynamic systems.
- Recurrent neural networks and LSTM architectures for nonlinear dynamics.
- Practical implementation of time series forecasting in Python.

- A reactor-based case study illustrating engineering relevance.

In the next chapter, we extend these ideas to Model Validation, Uncertainty, and Interpretability.

9 Chapter 9: Model Validation, Uncertainty, and Interpretability

9.1 Introduction

Machine learning models are increasingly used to support decision-making in chemical engineering applications, including process optimization, fault detection, and yield prediction. However, a model that performs well on historical data does not necessarily provide reliable predictions in real operating conditions.

Model validation and uncertainty quantification are therefore critical. In chemical processes, incorrect predictions can lead to unsafe operation, economic losses, or regulatory non-compliance. In addition, engineers must be able to understand why a model makes a certain prediction, particularly when models are used to guide operational or design decisions.

This chapter introduces the principles of model validation, uncertainty assessment, and interpretability, emphasizing their importance in trustworthy and deployable machine learning models for chemical engineering.

9.2 Conceptual Foundations

9.2.1 Overfitting and Underfitting

A fundamental challenge in model development is balancing model complexity and generalization.

- **Underfitting** occurs when a model is too simple to capture the underlying relationship between inputs and outputs.
- **Overfitting** occurs when a model fits noise in the training data rather than the true process behavior.

Mathematically, the expected prediction error can be decomposed into bias and variance:

$$E[(y - \hat{y})^2] = Bias^2 + Variance + Noise$$

Chemical engineering datasets often contain noise from sensors, disturbances, and unmeasured variables, making overfitting a serious concern.

9.2.2 Cross-Validation

Cross-validation provides a systematic way to estimate a model's ability to generalize to unseen data.

In k -fold cross-validation, the dataset is divided into k subsets. The model is trained on $k - 1$ folds and evaluated on the remaining fold. This process is repeated k times, and performance metrics are averaged:

$$CV\ Error = \frac{1}{k} \sum_{i=1}^k E_i$$

where E_i is the error on the i -th validation fold.

Cross-validation is particularly important in chemical engineering, where datasets may be small or expensive to generate.

9.2.3 Model Interpretability

Interpretability refers to the ability to explain how input variables influence model predictions.

Interpretability is critical in process engineering because:

- Models must align with physical intuition.
- Engineers need to verify that predictions are physically reasonable.
- Regulatory and safety requirements demand transparency.

Common interpretability approaches include:

- Feature importance analysis
- Partial dependence plots
- Sensitivity analysis

Interpretable models allow engineers to assess whether a model has learned meaningful process relationships or spurious correlations.

9.3 Python Implementation

9.3.1 Evaluating Regression and Classification Models

We demonstrate validation techniques using synthetic process data.

```

import numpy as np
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.metrics import mean_squared_error, accuracy_score

## Regression Model Validation

np.random.seed(0)

X = np.random.rand(200, 3)
y = 10 + 5*X[:,0] - 3*X[:,1] + np.random.normal(0, 0.5, size=200)

model = LinearRegression()

cv_mse = -cross_val_score(
    model, X, y, cv=5, scoring="neg_mean_squared_error"
)

cv_mse.mean()

```

0.2569785817465236

Cross-validation provides a robust estimate of prediction error, reducing reliance on a single train-test split.

```

# Classification Model Validation
X_class = np.random.rand(200, 2)
y_class = (X_class[:,0] + X_class[:,1] > 1).astype(int)

clf = LogisticRegression()
clf.fit(X_class, y_class)

y_pred = clf.predict(X_class)
accuracy_score(y_class, y_pred)

```

0.965

For classification tasks, accuracy, precision, recall, and confusion matrices should be evaluated together to avoid misleading conclusions.

9.4 Engineering Case Study

9.4.1 Quantifying Uncertainty in Predicted Reactor Yields

Consider a reactor where product yield depends on temperature, residence time, and feed composition. A regression model is trained to predict yield, but engineers must understand the confidence in

these predictions.

After fitting a regression model, uncertainty can be assessed by analyzing prediction residuals and their variance:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Higher uncertainty may indicate:

- Insufficient data in certain operating regions
- Missing input variables
- Regime changes in process behavior

Quantifying uncertainty allows engineers to:

- Define safety margins
- Identify unreliable predictions
- Prioritize additional data collection

9.5 Chapter Summary

This chapter emphasized the importance of validating and interpreting machine learning models in chemical engineering applications. Key topics included:

- The risks of overfitting and underfitting
- Cross-validation as a robust performance evaluation tool
- The need for interpretability in engineering decision-making
- Practical validation of regression and classification models
- Quantifying uncertainty in reactor yield predictions

Robust validation and transparent models are essential for deploying machine learning safely and effectively in real chemical processes.

10 Chapter 10: Integrating Machine Learning into Chemical Engineering Practice

10.1 Introduction

Machine learning models achieve their true value only when they are successfully integrated into real chemical engineering workflows. While previous chapters focused on model development, validation, and interpretation, practical deployment introduces additional challenges that are often overlooked in purely academic treatments.

Chemical processes operate continuously, under changing conditions, and within strict safety and economic constraints. Models must therefore be reliable, maintainable, and aligned with engineering judgment. This chapter bridges the gap between theoretical model development and real-world chemical engineering practice, emphasizing how machine learning models are deployed, monitored, and maintained over their operational lifetime.

10.2 Conceptual Foundations

10.2.1 Deployment Strategies for Machine Learning Models

Deployment refers to the process of embedding a trained machine learning model into an operational environment where it can generate predictions using live or periodically updated data.

From an engineering perspective, deployment strategies typically fall into three categories:

- 1- Offline decision support
- 2- Online advisory systems
- 3- Closed-loop integration

In offline decision support, models are used for scenario analysis, optimization studies, or planning tasks. Predictions do not directly affect process operation but inform engineering decisions.

Online advisory systems provide real-time predictions or recommendations to operators. The final decision remains with human experts, preserving a safety layer between the model and the process.

Closed-loop integration embeds the model directly into control or optimization algorithms. This approach requires extensive validation, redundancy, and safeguards due to its direct impact on process behavior.

10.2.2 Monitoring Model Performance

Once deployed, a machine learning model must be continuously monitored to ensure its predictions remain accurate and physically meaningful.

Key performance indicators include:

- Prediction error metrics over time
- Distribution shifts in input variables
- Frequency of extrapolation beyond training data

In chemical processes, changes in catalyst activity, feedstock quality, equipment fouling, or sensor calibration can cause data drift, leading to degraded model performance.

Mathematically, data drift can be detected by comparing statistical properties of input variables:

$$\Delta\mu = \mu_{current} - \mu_{training}$$

$$\Delta\sigma^2 = \sigma_{current}^2 - \sigma_{training}^2$$

Significant deviations may indicate that the model is being used outside its valid operating envelope.

10.2.3 Model Maintenance and Lifecycle Management

Machine learning models are not static assets. They require periodic retraining, recalibration, and, in some cases, redesign.

Maintenance strategies include:

- Scheduled retraining using updated datasets
- Incremental learning for slowly evolving processes
- Model retirement when assumptions are no longer valid

From an engineering standpoint, every model must have:

- A clearly defined domain of applicability
- Documented assumptions and limitations
- Traceability between data, model versions, and decisions

This lifecycle-oriented view aligns machine learning practice with traditional chemical engineering standards for equipment, control systems, and safety analysis.

10.3 Engineering Case Study

10.3.1 Decision Support for Process Optimization

Consider a chemical reactor where yield depends on temperature, pressure, and residence time. A machine learning regression model has been developed to predict yield across feasible operating conditions.

The model is deployed as an offline decision support tool to assist engineers in identifying optimal operating points under varying economic constraints. The trained model is used to evaluate yield predictions across a grid of operating conditions. Engineers then apply physical constraints, safety limits, and economic considerations to select feasible operating points.

Importantly, the model does not replace engineering judgment. Instead, it narrows the search space and provides quantitative insight into trade-offs between competing objectives.

10.4 Chapter Summary

This chapter addressed the practical integration of machine learning into chemical engineering workflows. Key points included:

- Deployment strategies ranging from offline analysis to closed-loop integration
- The necessity of monitoring model performance under changing process conditions
- Lifecycle management of machine learning models as engineering assets
- A decision support case study illustrating responsible use of ML for process optimization

With this chapter, the book completes the transition from foundational theory to applied machine learning practice in chemical engineering, preparing readers to deploy models responsibly in real industrial environments.

11 Appendix A: Python Primer for Engineers

Python has become the dominant programming language for data-driven modeling due to its readability, extensive scientific ecosystem, and suitability for rapid prototyping. This appendix provides a concise reference for Python concepts most relevant to chemical engineers using machine learning.

11.1 Variables and Data Types

Python uses dynamic typing, but understanding data types is essential for numerical computing:

- int: Integer values
- float: Floating-point numbers
- bool: Logical values
- list: Ordered collections
- tuple: Immutable ordered collections
- dict: Key–value mappings

```
temperature = 600.0 #float
pressure = 20 #int
is_safe = True #bool
```

11.1.1 Control Structures

Conditional statements and loops allow logical flow control in engineering computations.

```
if temperature > 650:
    status = "High temperature"
else:
    status = "Normal operation"

#looping
for i in range(5):
    print(i)
```

11.1.2 Functions and Modularity

Functions encapsulate reusable logic and support clean model developmen

```
def reactor_yield(T, P):  
    return 0.5 + 0.001*T + 0.01*P
```

Modular code improves reproducibility, maintainability, and traceability—key requirements in engineering practice.

12 Appendix B: NumPy, Pandas, scikit-learn, and TensorFlow Cheat Sheet

This appendix summarizes the most frequently used commands across the core machine learning stack. `## NumPy`

Efficient numerical computation and array manipulation.

```
import numpy as np

x = np.array([1, 2, 3])
x.mean()
np.linalg.inv(np.eye(3))
```

Key uses in chemical engineering:

- Vectorized calculations
- Linear algebra
- Synthetic data generation

12.1 Pandas

Structured data handling using tabular formats. `“ import pandas as pd`

```
df = pd.DataFrame({ "Temperature": [550, 600, 650], "Yield": [0.7, 0.75, 0.78] })
```

```
df.describe()
```

`““ Used extensively for:`

- Process datasets
- Experimental records
- Data preprocessing

12.2 scikit-learn

Classical machine learning models and validation tools.

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
```

Core functionalities:

Regression and classification models

Clustering and PCA

Cross-validation and metrics

12.2.1 TensorFlow

Neural network modeling for nonlinear systems.

```
import tensorflow as tf
from tensorflow.keras import layers
```

Primarily used when

- Strong nonlinearities exist
- High-dimensional input–output mappings are required

13 Appendix C: Synthetic Data Generation for Chemical Processes

In many industrial and academic settings, real process data may be unavailable due to confidentiality, cost, or safety constraints. Synthetic data provides a controlled and physically interpretable alternative.

13.1 Principles of Synthetic Data Generation

Synthetic datasets should:

- Reflect known physical relationships
- Include realistic noise
- Respect operating constraints

A general formulation is:

$$y = f(x_1, x_2, \dots, x_n) + \varepsilon$$

where ε represents measurement noise or unmodeled effects. **Example: Synthetic Reactor Data**

```
import numpy as np

T = np.random.uniform(500, 700, 500)
P = np.random.uniform(10, 30, 500)

yield_data = (
    0.6
    + 0.0004 * (T - 600)
    + 0.01 * (P - 20)
    + np.random.normal(0, 0.02, 500)
)
```

Synthetic data enables:

- Method development
- Sensitivity analysis
- Controlled benchmarking of models

14 Appendix D: Recommended Further Reading

The following references provide authoritative coverage of machine learning, data analysis, and modeling in chemical engineering contexts.

14.1 Machine Learning and Data Science

- Bishop, C. M., Pattern Recognition and Machine Learning
- Hastie, T., Tibshirani, R., Friedman, J., The Elements of Statistical Learning
- Murphy, K. P., Machine Learning: A Probabilistic Perspective

14.2 Chemical Engineering Applications

- Stephanopoulos, G., Chemical Process Control
- Ogunnaike, B. A., Ray, W. H., Process Dynamics, Modeling, and Control
- Marlin, T. E., Process Control: Designing Processes and Control Systems

14.3 Data-Driven Process Modeling

- Qin, S. J., “Statistical Process Monitoring”
- Kadlec, P., Gabrys, B., Strandt, S., “Data-driven Soft Sensors”

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